Equational Reasoning in ∞-Categories

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ABSTRACT

We present a compact typed equational theory for ∞-categories. In its strong reading, where equality is identity, the theory axiomatizes strict ω-categories. We show that models of the theory are equivalent to strict ω-categories and that free models exist by a universal Horn construction. For the free model we give an effective decision procedure for equality of closed terms by deterministic normalization, making the framework amenable to automation and rewriting. The same syntax also supports a weak reading in which equality is interpreted as homotopy relative to boundaries. We give a constructive homotopical semantics in which n-cells are boundary-preserving homotopies and equations are witnessed by higher cells, so that the structural laws hold up to coherent equivalence. We develop the proof calculus for typing, composition, exchange, and identities, establish initiality and soundness results. The result is a single algebraic framework that connects strict higher categories with homotopy practice.

CCS CONCEPTS

• **Theory of computation → Type theory**; **Constructive mathematics**; **Equational logic and rewriting**; **Automated reasoning**; **Logic and verification**.

KEYWORDS

type theory, constructive mathematics, equational logic, automated reasoning, higher categories, homotopy

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1 INTRODUCTION

Equational reasoning is a central method in ordinary category theory: one reasons about composites by writing equations between terms built from identities, composition, and functorial structure. The modern theory of higher categories and ∞-categories aspires to similar clarity, but it must account for infinitely many levels of cells and for coherence of composition across these levels. There are now several mature and equivalent models of $(\infty, 1)$-categories, including quasicategories, complete Segal spaces, and simplicial categories [3, 4, 9, 17], and there are extensive treatments and applications in topology, algebra, and geometry [7, 12–14]. These accounts are mathematically powerful, but their definitions are often technically involved, and routine equalities are often proved inside a chosen model using model specific infrastructure (i.e., combinatorial or homotopical conditions). This makes it harder to present everyday higher categorical calculations as short symbolic derivations, and it raises a real barrier for mechanization and automation. Existing algebraic presentations of strict higher categories and homotopical models of $(\infty, 1)$-categories, e.g. [3, 4, 7, 9, 12–14, 17], provide robust foundations. Equational proofs in the homotopical setting are typically internal to a chosen model, while algebraic treatments of strictness seldom supply a decidable proof system.

Our starting point is the observation that many basic constructions in higher categories can be presented as *equational judgments*. For a 1-cell $f : a \to b$, we can write its type as the pair of equations $\mathsf{src}\, f = a$ and $\mathsf{tgt}\, f = b$. For a 2-cell $\alpha : f \Rightarrow g$, we can write $\mathsf{src}\, \alpha = f$ and $\mathsf{tgt}\, \alpha = g$, and so on. Vertical and horizontal composition become polymorphic operations subject to typing judgments, associativity, and the exchange law. Identities are given by a uniform operator that introduces a higher cell with specified source and target. This equational reading is intuitive, mirroring ordinary categorical practice and suggesting a compact first-order axiomatization.

We develop this idea as a typed equational theory indexed by the natural numbers. The signature contains sorts for n-cells, source and target operators src_{ij} and tgt_{ij}, a family of polymorphic compositions $\circ_n$, and identity operators id_{ij}. The axioms are universal Horn formulas that encode typing, associativity at every level, exchange between different levels of composition, and identity laws. This presentation supports a canonical free construction from well-typed terms modulo provable equality. A pragmatic motivation for analyzing the free model is mechanization. Even when the intended mathematics is about composition, mechanized proofs are often dominated by boundary bookkeeping: equalities built from iterated src/tgt, identities, and sources and targets of composites arise constantly, and they tend to accumulate into brittle collections of ad hoc simplification lemmas and rewriting heuristics. We provide a deterministic normalization procedure, yielding an effective decision procedure for equality of closed terms in the free model. In particular, normalization reduces these bookkeeping goals to comparison of normal forms, providing a small and stable kernel for automation of higher-dimensional "plumbing" in the free setting. In contrast, we show an undecidability result for the general model, showing that it is undecidable to determine if two terms, e_1 and e_2, are either both undefined or both defined and equal. This is done by by encoding the word problem for finitely presented semigroups into definitional equality.

The same equational theory admits two readings. In the *strong* reading, equality is identity. In this mode we prove an adequacy theorem: models of the strong theory are exactly strict ω categories. In the *weak* reading, equality is interpreted as an equivalence in

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which equations are inhabited by higher cells. Concretely, we give an explicit homotopy model over topological spaces in which cells are boundary preserving homotopies. This produces an equational framework for $(\infty, 1)$-style reasoning without committing to a single ambient model. Our weak reading is compatible with established homotopy models of $(\infty, 1)$-categories [3, 4, 9, 17]. We further identify a small family of explicit such homotopies of a restricted form, so-called *reparameterizations*, that are sufficient to witness the structural laws we use in practice, including associativity, identities, and exchange. This yields a direct route from symbolic derivations to explicit coherence data.

The contribution here is to provide an equational calculus that covers both homotopical equalities and the strict case, so one can switch between strict and weak interpretations without changing signatures. This supports formal development and mechanization and creates a direct route from equational derivations in the strong mode to homotopical statements in the weak mode.

Contributions. The paper makes the following contributions.

- *A compact Horn axiomatization.* We give a single typed equational theory for higher composition, identities, and exchange.
- *Algebraic adequacy for the strong reading.* We prove that models of the strong theory are equivalent to strict ω-categories, and we construct free models as quotients of well-typed terms modulo provable equations.
- *A decision procedure.* We give a normalization-based decision procedure that decides well-formedness of terms and derivability of equations between ground terms in the free model.
- *A weak homotopical mode.* We define a weak reading in which equality is interpreted as homotopy relative to boundary and develop the associated calculus. We show that the basic structural rules are validated in homotopy models and formulate a general soundness theorem.

Outline. §2.2 develops the general calculus of higher composition and records the type and equality invariants that we use throughout. §3 presents the universal Horn axiomatization and its free model. §4 gives the decision procedure for the free model, and §5 proves undecidability in the general case. §6 proves algebraic adequacy for the strong reading with respect to strict ω categories. §7 introduces the weak reading, defines homotopy relative to boundary, and proves soundness of the basic structural rules. §8 gives a general soundness theorem. Then, §9 concludes.

Some proofs omitted from the main text can be found in the appendix with hyperlinks from the main text.

2 THE EQUATIONAL CALCULUS

We view an ∞-category as a model of a typed equational theory of terms and equations (given in §3). The theory is presented by a set of rules for deriving types of expressions and equations between expressions. This is the *strong* version of the theory, in which equations are identities. There is also a *weak* version in which equations are equivalences. For background on equational logic see [11, 15].

$$\begin{tikzcd}[column sep=huge] \operatorname{src}\operatorname{src}\alpha = \operatorname{src}\operatorname{tgt}\alpha \arrow[r, bend left, "\operatorname{src}\alpha"] \arrow[r, bend right, "\operatorname{tgt}\alpha"'] \arrow[r, phantom, "\Downarrow \alpha"] & \operatorname{tgt}\operatorname{src}\alpha = \operatorname{tgt}\operatorname{tgt}\alpha \end{tikzcd}$$

(a) Globular reading

$$\begin{tikzcd}[column sep=huge] a \arrow[r, bend left, "f : a \to b"] \arrow[r, bend right, "g : a \to b"'] \arrow[r, phantom, "\Downarrow\, \alpha : f \to g : a \to b"] & b \end{tikzcd}$$

(b) Typed reading

Figure 1: Two equivalent readings of the same cell equation.

2.1 Globular Data and Typing

An ∞-category is specified by a collection of pairwise disjoint nonempty sets C_n for each $n \geq 0$ with source and target maps $\operatorname{src}, \operatorname{tgt} : C_{n+1} \to C_n$ specifying a functional type for each element and satisfying the *globularity axioms*, that is, for all $n \geq 2$ and $\alpha \in C_n$, $\operatorname{src}\operatorname{src}\alpha = \operatorname{src}\operatorname{tgt}\alpha$ and $\operatorname{tgt}\operatorname{src}\alpha = \operatorname{tgt}\operatorname{tgt}\alpha$. From the globularity axioms it follows that any composition of src and tgt of length n beginning with src is equal to src^n, and any such composition beginning with tgt is equal to tgt^n; for example, if $\dim\alpha \geq 5$, then $\operatorname{src}\operatorname{src}\operatorname{tgt}\operatorname{src}\operatorname{tgt}\alpha = \operatorname{src}\operatorname{src}\operatorname{src}\operatorname{src}\operatorname{src}\alpha = \operatorname{src}^5 \alpha$.

The elements of C_n are called *cells of dimension n* or *n-cells*. The dimension of f is denoted $\dim f$. For $\dim f \geq 1$, we often write $f : a \to b$ to mean $\operatorname{src} f = a$ and $\operatorname{tgt} f = b$, and for $\dim\alpha \geq 2$, we often write $\alpha : f \to g : a \to b$ to mean $\operatorname{src}\alpha = f$, $\operatorname{tgt}\alpha = g$, $\operatorname{src} f = \operatorname{src} g = a$, $\operatorname{tgt} f = \operatorname{tgt} g = b$. With this notation, Figures 1(a) and 1(b) have the same meaning.

2.2 Composition and Identities

The composition operator $\circ_k \colon C_n \times C_n \to C_n$ may be applied only to cells of dimension $n > k$. It allows two n-cells to be composed through a common k-dimensional subcell. As mentioned above, for $n = \dim\alpha = \dim\beta$ and $0 \leq k < n$, if $\operatorname{src}^k \beta = \operatorname{tgt}^k \alpha$, then $\beta \circ_{n-k} \alpha$ is well formed and $\dim(\beta \circ_{n-k} \alpha) = n$.

It will follow from our equational axioms below that for $k > 1$,

$$\begin{aligned} \operatorname{src}(\beta \circ_{n-1} \alpha) &= \operatorname{src}\alpha & \qquad \operatorname{src}(\beta \circ_{n-k} \alpha) &= \operatorname{src}\beta \circ_{n-k} \operatorname{src}\alpha \\ \operatorname{tgt}(\beta \circ_{n-1} \alpha) &= \operatorname{tgt}\beta & \qquad \operatorname{tgt}(\beta \circ_{n-k} \alpha) &= \operatorname{tgt}\beta \circ_{n-k} \operatorname{tgt}\alpha. \end{aligned}$$

Thus $\operatorname{src}^k(\beta \circ_{n-k} \alpha) = \operatorname{src}^k \alpha$ and $\operatorname{tgt}^k(\beta \circ_{n-k} \alpha) = \operatorname{tgt}^k \beta$. These inferences correspond to the typing rules

$$\frac{f : a \to b \qquad g : b \to c}{g \circ_{n-1} f : a \to c} \tag{1}$$

$$\frac{\alpha : f \to g \qquad \beta : h \to \ell \qquad h \circ_{n-k} f : a \to c \qquad \ell \circ_{n-k} g : a \to c}{\beta \circ_{n-k} \alpha : h \circ_{n-k} f \to \ell \circ_{n-k} g} \tag{2}$$

The rule (2) is illustrated below.

$$\begin{tikzcd} a \arrow[r, bend left=60, "f"] \arrow[r, bend right=60, "g"'] \arrow[r, phantom, "\Downarrow\alpha"] & b \arrow[r, bend left=60, "h"] \arrow[r, bend right=60, "\ell"'] \arrow[r, phantom, "\Downarrow\beta"] & c \end{tikzcd}$$

All composition operators are postulated to be associative in the sense that

$$(\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha = \gamma \circ_{n-k} (\beta \circ_{n-k} \alpha).$$

There is also a unary operator id with $\dim \mathrm{id}\, a = \dim a + 1$ that gives gives two-sided identities for composition. We postulate that id satisfies the following equations, where $n = \dim \alpha$:

$$\mathrm{src}\, \mathrm{id}\, a = \mathrm{tgt}\, \mathrm{id}\, a = a \qquad \alpha \circ_{n-k} \mathrm{id}^k\, \mathrm{src}^k\, \alpha = \alpha \qquad \mathrm{id}^k\, \mathrm{tgt}^k\, \alpha \circ_{n-k} \alpha = \alpha$$

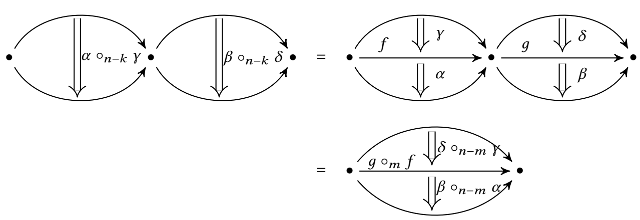
It follows from the first two equations that $\mathrm{id}\, a : a \to a$.

We also postulate that identities compose to give identities in all meaningful ways, as axiomatized by the following rule for $n = \dim f = \dim g$ and $1 \leq k < n$: if $\mathrm{tgt}^k f = \mathrm{src}^k g$, so that $g \circ_{n-k} f$ is well formed, then $\mathrm{tgt}^{k+1}\, \mathrm{id}\, f = \mathrm{src}^{k+1}\, \mathrm{id}\, g$, so that $\mathrm{id}\, g \circ_{n-k} \mathrm{id}\, f$ is also well formed, and

$$\mathrm{id}(g \circ_{n-k} f) = \mathrm{id}\, g \circ_{n-k} \mathrm{id}\, f. \tag{3}$$

Any pair of composition operators $\circ_m$ and $\circ_k$ with $m \neq k$ are related by an exchange law. If $n = \dim \alpha = \dim \beta = \dim \gamma = \dim \delta$ and $1 \leq k < m < n$, and if the appropriate boundary conditions hold (to wit, $b = \mathrm{tgt}^m \gamma = \mathrm{tgt}^m \alpha = \mathrm{src}^m \delta = \mathrm{src}^m \beta$, $f = \mathrm{tgt}^k \gamma = \mathrm{src}^k \alpha$, $g = \mathrm{tgt}^k \delta = \mathrm{src}^k \beta$, and $g \circ_{n-m} f = \mathrm{tgt}^k(\delta \circ_{n-m} \gamma) = \mathrm{src}^k(\beta \circ_{n-m} \alpha)$), then

$$(\beta \circ_{n-m} \alpha) \circ_{n-k} (\delta \circ_{n-m} \gamma) = (\beta \circ_{n-k} \delta) \circ_{n-m} (\alpha \circ_{n-k} \gamma).$$



2.3 $(\infty, 1)$-Categories

In homotopy theory, particularly in the context of homotopy type theory, $(\infty, 1)$−categories are essential for modeling spaces and their higher homotopies. An $(\infty, 1)$-category is an ∞-category in which all n-cells for $n \geq 2$ are invertible with respect to sequential composition $\circ_1$. This provides a framework for modeling spaces up to homotopy, allowing for a natural modeling of higher-dimensional morphisms and equivalences. We can make this explicit in the language with an operator $(-)^{\leftarrow} : C_{n+1} \to C_{n+1}$ axiomatized by

$$\frac{f : a \to b}{f^{\leftarrow} : b \to a} \qquad \frac{f : a \to b}{f^{\leftarrow} \circ f = \mathrm{id}\, a} \qquad \frac{f : a \to b}{f \circ f^{\leftarrow} = \mathrm{id}\, b}$$

Having an explicit inverse operator entails some loss of generality, as it does not follow from the axioms for ∞-categories that inverses exist. However, two-sided inverses are unique when they do exist. We also have a derived rule

$$\frac{f : a \to b}{f^{\leftarrow\leftarrow} = f}$$

2.4 ∞-Functors and ∞-Natural Transformations

If C and D are ∞-categories, an ∞-functor $F : C \to D$ is a map $F : C \to D$ such that

- F preserves dimension; that is, $\dim(Fa) = \dim a$;
- F commutes with the source and target maps; that is, $\mathrm{src}(Ff) = F(\mathrm{src}\, f)$ and $\mathrm{tgt}(Ff) = F(\mathrm{tgt}\, f)$. Equivalently, the following typing rule is sound:

$$\frac{f : a \to b}{Ff : Fa \to Fb}$$

- F respects composition: $F(g \circ_k f) = Fg \circ_k Ff$;
- F respects identities: $F\, \mathrm{id}\, a = \mathrm{id}\, Fa$.

An $(\infty, 1)$-functor is an ∞-functor between $(\infty, 1)$-categories. One can show that $(\infty, 1)$-functors also preserve inverses: if $\alpha : f \to g$, then $\alpha^{\leftarrow} : g \to f$, $F(\alpha^{\leftarrow}) : Fg \to Ff$, $F\alpha : Ff \to Fg$, $(F\alpha)^{\leftarrow} : Fg \to Ff$, and

$$\begin{aligned} F(\alpha^{\leftarrow}) &= F(\alpha^{\leftarrow}) \circ \mathrm{id}\, Fg = F(\alpha^{\leftarrow}) \circ (F(\alpha) \circ (F\alpha)^{\leftarrow}) \\ &= (F(\alpha^{\leftarrow}) \circ F(\alpha)) \circ (F\alpha)^{\leftarrow} = F(\alpha^{\leftarrow} \circ \alpha) \circ (F\alpha)^{\leftarrow} \\ &= F\, \mathrm{id}\, f \circ (F\alpha)^{\leftarrow} = \mathrm{id}\, Ff \circ (F\alpha)^{\leftarrow} = (F\alpha)^{\leftarrow}. \end{aligned}$$

If C and D are ∞-categories and $F, G : C \to D$ are ∞-functors, an ∞-natural transformation $\varphi : F \to G : C \to D$ is a function $\varphi : C \to D$ such that for all a, b, and $f : a \to b$ in C,

$$\varphi_a : Fa \to Ga \qquad Gf \circ_{n-1} \varphi_a = \varphi_b \circ_{n-1} Ff.$$

3 UNIVERSAL HORN AXIOMATIZATION

We collect all the axioms into an equational theory axiomatized by universal Horn formulas. This makes precise what it means for a composite term to be well-formed and provides free models by a standard term model construction.

3.1 The Theory EqCat_∞

3.1.1 Syntax. The language of our universal Horn axiomatization consists of a set of variables X, each with a specified dimension [1] $\dim : X \to \mathbb{N}$, unary function symbols src, tgt, and id, and binary function symbols $\circ_n$ for all $n \in \mathbb{N}$. From these primitives we can construct a set of well-formed terms and equations, each with an inductively defined dimension, according to the following rules.

- Each variable $x \in X$ is well-formed, and its dimension $\dim x \in \mathbb{N}$ is given *a priori*.
- If e is well-formed and $\dim e = n$, then $\mathrm{src}\, e$ and $\mathrm{tgt}\, e$ are well-formed and $\dim \mathrm{src}\, e = \dim \mathrm{tgt}\, e = n - 1$.
- If e is well-formed and $\dim e = i$, then $\mathrm{id}\, e$ is well-formed and $\dim \mathrm{id}\, e = i + 1$.

[1] These axioms as stated assuming that dimensions are natural numbers with their usual order. In fact, our arguments go through just as easily when the dimensions are drawn from any tree-ordered set (partial order with bottom element where the predecessors of any element are linearly-ordered); an axiomatization allowing for such generalized dimensions is given in the appendix.

Dimension axioms

(D1) $\dim x > 0 \Rightarrow \dim \operatorname{src} x = \dim x - 1$
(D2) $\dim x > 0 \Rightarrow \dim \operatorname{tgt} x = \dim x - 1$
(D3) $d = \dim x = \dim y > n \Rightarrow \dim(y \circ_n x) = d$
(D4) $\dim \operatorname{id} x = \dim x + 1$

Equality axioms, structural

(E1) $x = x$
(E2) $x = y \Rightarrow y = x$
(E3) $x = y \Rightarrow y = z \Rightarrow x = z$
(E4) $x = y \Rightarrow \dim x = \dim y$
(E5) $x = y \Rightarrow \dim x = \dim y > 0 \Rightarrow \operatorname{src} x = \operatorname{src} y$
(E6) $x = y \Rightarrow \dim x = \dim y > 0 \Rightarrow \operatorname{tgt} x = \operatorname{tgt} y$
(E7) $x = y \Rightarrow z = w \Rightarrow \dim x = \dim y = \dim z = \dim w > n \Rightarrow x \circ_n z = y \circ_n w$
(E8) $x = y \Rightarrow \operatorname{id} x = \operatorname{id} y$

Equality axioms, ∞-categorical

(IC1) $\dim x \geq 2 \Rightarrow \operatorname{src}\operatorname{src} x = \operatorname{src}\operatorname{tgt} x$
(IC2) $\dim x \geq 2 \Rightarrow \operatorname{tgt}\operatorname{src} x = \operatorname{tgt}\operatorname{tgt} x$
(IC3) $\operatorname{src}\operatorname{id} x = x$
(IC4) $\operatorname{tgt}\operatorname{id} x = x$
(IC5) $\dim x = \dim y = n+1 \Rightarrow \operatorname{src}(y \circ_n x) = \operatorname{src} x$
(IC6) $\dim x = \dim y = n+1 \Rightarrow \operatorname{tgt}(y \circ_n x) = \operatorname{tgt} y$
(IC7) $\dim x = \dim y > n+1 \Rightarrow \operatorname{src}(y \circ_n x) = \operatorname{src} y \circ_n \operatorname{src} x$
(IC8) $\dim x = \dim y > n+1 \Rightarrow \operatorname{tgt}(y \circ_n x) = \operatorname{tgt} y \circ_n \operatorname{tgt} x$
(IC9) $\dim x = \dim y = \dim z > n \Rightarrow (x \circ_n y) \circ_n z = x \circ_n (y \circ_n z)$
(IC10) $\dim x = \dim y = \dim z = \dim w > \max m, n \Rightarrow (x \circ_n y) \circ_m (z \circ_n w) = (x \circ_m z) \circ_n (y \circ_m w), m \neq n$
(IC11) $d = \dim x > n \Rightarrow x \circ_n \operatorname{id}^{d-n} \operatorname{src}^{d-n} x = x$
(IC12) $d = \dim x > n \Rightarrow \operatorname{id}^{d-n} \operatorname{tgt}^{d-n} x \circ_n x = x$
(IC13) $d = \dim x = \dim y > n \Rightarrow \operatorname{src}^{d-n} y = \operatorname{tgt}^{d-n} x \Rightarrow \operatorname{id}(y \circ_n x) = \operatorname{id} y \circ_n \operatorname{id} x$

Figure 2: EqCat_∞: a universal Horn axiomatization for ∞-categories.

- If e and e' are well-formed and $\dim e = \dim e' = n$, then $e \circ_k e'$ is well-formed and $\dim(e \circ_k e') = n$.
- If e and e' are well-formed and $\dim e = \dim e' = n$, then $e = e'$ is well-formed and $\dim(e = e') = n$.

3.1.2 Semantics. Well-formed terms are not necessarily meaningful under all interpretations. That is because there is an extra equality premise $\operatorname{src}^k y = \operatorname{tgt}^k x$ that must be satisfied in order that a composition $y \circ_{n-k} x$ be meaningful, where $\dim y = \dim x = n$. The axioms listed in Fig. 2 along with this implication determine inductively when a term is meaningful and whether two meaningful terms are equal.

Let C be a generalized ∞-category. An *interpretation* is specified by a dimension-preserving function $\sigma : X \to C$, that is, $\dim(\sigma(x)) = \dim x$ for all $x \in X$. Each such σ extends inductively to a *partial* function σ : {well-formed terms} $\rightharpoonup C$ preserving dimensions according to the following rules. We will say that a term e is *defined* (under the interpretation σ) if it is in the domain of σ.

- Each variable $x \in X$ is defined, and its value is $\sigma(x)$.
- If e is defined and $\dim e = m$, then $\operatorname{src} e$ and $\operatorname{tgt} e$ are defined, and $\sigma(\operatorname{src} e) = \operatorname{src}(\sigma(e))$ and $\sigma(\operatorname{tgt} e) = \operatorname{tgt}(\sigma(e))$.
- If e is defined and $\dim e = i$, then $\operatorname{id} e$ is defined, and $\sigma(\operatorname{id} e) = \operatorname{id}(\sigma(e))$.
- If e and e' are defined and $\dim e = \dim e' = n$, and $\operatorname{src}^k(\sigma(e)) = \operatorname{tgt}^k(\sigma(e'))$, then $e \circ_{n-k} e'$ is defined and $\sigma(e \circ_{n-k} e') = \sigma(e) \circ_{n-k} \sigma(e')$.
- If e and e' are defined and $\dim e = \dim e' = n$, then $e = e'$ is defined and is true iff $\sigma(e) = \sigma(e')$.

The equational axioms are given in Fig. 2. The axioms for equality are divided into the standard axioms for equality and those specific to (generalized) ∞-categories. The standard axioms are reflexivity, symmetry, transitivity, and congruence with respect to all operations. Our deductive system consists of equational Horn logic. Axioms IC1 and IC2 are the globularity axioms that ensure that source and target maps commute appropriately across dimensions, as in the standard definition of globular sets [13, Section 1.1] or [2].

The definition of well-formedness in §3.1.1 corresponds to the dimension axioms (D1)-(D4). These can be checked purely syntactically, independent of any interpretation.

3.2 Free Models

Having a universal Horn axiomatization means that there are free models that can be constructed as quotients of certain well-formed terms modulo provable equations. The free construction is left adjoint to the forgetful functor from ∞-categories to *stratified sets* consisting of pairs $(C, \dim)$, where C is a set and $\dim : C \to \mathbb{N}$.

Given a stratified set $(C, \dim)$, the free construction generates an ∞-category consisting of certain well-formed terms over primitive symbols C modulo provable equations. The dimension of each term is determined by axioms (D1)-(D4). Not all well-formed terms will be included, but only those that are meaningful in the free model. This may seem like a circular statement, but it will follow from our normalization procedure in §4 that meaningfulness in the free model is decidable and inductively determined; moreover, it will suffice to consider only terms in normal form.

The corresponding forgetful functor maps an ∞-category on cells C to its underlying stratified set $(C, \dim)$, forgetting all algebraic structure except dimension.

4 DECIDABILITY IN THE FREE MODEL

Let C be a stratified set, which is to say a set C together with an assignment dim of natural-number dimensions to elements of C. For us the free model generated by C is the result of applying the adjoint of the forgetful functor from the category of (small) ∞-categories to the category of stratified sets to C. In particular, expressions in the language of ∞-categories with constants from C are only well-formed if this is forced by their dimensions. We do not allow assuming any compatibility between pairs of generators of the same dimension, as is common practice when discussing free 1-categories generated by a labelled directed multigraph of generators, so in particular if f and g are 1-cells from a stratified set C then $g \circ_1 f$ is necessarily not a well-formed expression in the free ∞-category generated by C. In this section we show that equivalence in the free model is efficiently decidable. We do so by providing a deterministic *normalization algorithm* for well-formed terms. Given terms e_1, e_2, we compute their normal forms $\mathsf{NF}(e_1), \mathsf{NF}(e_2)$. The terms are equal in the free model if and only if both normalizations succeed and return the same normal form.

Definition 4.1 (Normal Form). A *normal form* is an expression of the shape

$$\mathsf{id}^\ell\, \mathsf{end}^m a$$

where $\ell, m \geq 0$, $\mathsf{end} \in \{\mathsf{src}, \mathsf{tgt}\}$, and a is a primitive symbol. We identify the two choices of end when $m = 0$ (so $\mathsf{id}^\ell\, \mathsf{src}^0\, a$ and $\mathsf{id}^\ell\, \mathsf{tgt}^0\, a$ represent the same normal form).

Lemma 4.2 (Equality of normal forms). *Let $u = \mathsf{id}^\ell\, \mathsf{end}_1^m a$ and $v = \mathsf{id}^i\, \mathsf{end}_2^j b$ be normal forms. Then $u = v$ in the free model iff*

$$a = b, \qquad \ell = i, \qquad m = j, \qquad \text{and if } m > 0 \text{ then } \mathsf{end}_1 = \mathsf{end}_2.$$

As mentioned previously, equivalence of two expressions and typeability of a single expression in a model are mutually dependent problems. We use the following axioms/theorems as rewrite rules in the left-to-right direction. Assume $\dim \alpha = \dim \beta = n$ and $k < n$. Then:

- (R1) $\mathsf{src}(\beta \circ_{n-1} \alpha) = \mathsf{src}\, \alpha$
- (R2) $\mathsf{tgt}(\beta \circ_{n-1} \alpha) = \mathsf{tgt}\, \beta$
- (R3) $\mathsf{src}(\beta \circ_k \alpha) = \mathsf{src}\, \beta \circ_k \mathsf{src}\, \alpha$
- (R4) $\mathsf{tgt}(\beta \circ_k \alpha) = \mathsf{tgt}\, \beta \circ_k \mathsf{tgt}\, \alpha,\ k \geq 2$
- (R5) $\mathsf{id}(\beta \circ_k \alpha) = \mathsf{id}\, \beta \circ_k \mathsf{id}\, \alpha$
- (R6) $\mathsf{src}\, \mathsf{id}\, \alpha = \alpha$
- (R7) $\mathsf{tgt}\, \mathsf{id}\, \alpha = \alpha$
- (R8) $\mathsf{src}\, \mathsf{tgt}\, \alpha = \mathsf{src}^2\, \alpha$
- (R9) $\mathsf{tgt}\, \mathsf{src}\, \alpha = \mathsf{tgt}^2\, \alpha$
- (R10) $\alpha \circ_{n-k} \mathsf{id}^k\, \mathsf{src}^k\, \alpha = \alpha$
- (R11) $\mathsf{id}^k\, \mathsf{tgt}^k\, \alpha \circ_{n-k} \alpha = \alpha$

Example 4.3. Fix a primitive symbol a with $\dim a = 3$. Let

$$u := \mathsf{id}^2\, \mathsf{tgt}^2\, a, \qquad v := \mathsf{id}^2\, \mathsf{tgt}\, \mathsf{src}\, a,$$
$$E := \big(\mathsf{id}^2\, \mathsf{tgt}^2(u \circ_1 v)\big) \circ_1 u.$$

Figure 3 shows the composition tree of E. In Steps 0–3 below we normalize E and obtain $\mathsf{NF}(E) = u$.

The normalization algorithm proceeds in three steps.

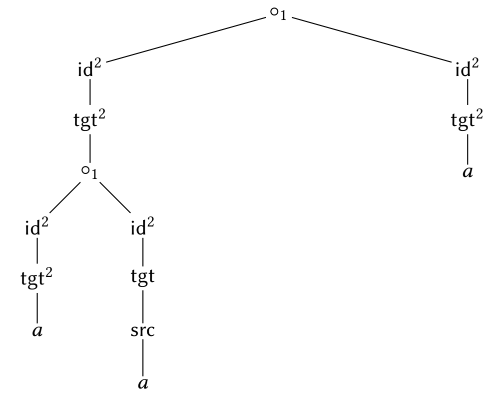


Figure 3: Composition tree for E from Example 4.3

Step 0: well formedness and dimension. To test the equivalence of two terms e_1 and e_2 in the free model, twe first check that e_1 and e_2 are well formed and that their dimensions match. Recall that a term over the operators $\mathsf{src}, \mathsf{tgt}, \mathsf{id}, \circ_k$ and primitive symbols $a, b, \ldots$ is *well formed* if every subterm has a dimension as calculated inductively according to the rules (D1)-(D4), where a primitive symbol a has dimension $\dim a$ as specified by the model.

Assuming that e_1 and e_2 are well formed, we apply the rules (R1)-(R11) above to simplify the expressions in the order specified below. If at any point we discover that a subterm is not typeable in the free model, then we halt and declare failure. Note that all the rules (R1)-(R11) preserve well-formedness.

Running example. The term E from Example 4.3 is well formed, and $\dim(E) = 3$.

Step 1: push unary operators below compositions. We exhaustively apply rules (R1)–(R5) to subterms. Applications of (R1) and (R2) eliminate a composition operator and applications of (R3)-(R5) move unary operators below composition operators, regarding terms as labeled trees with the root at the top. After this step, the term consists of a collection of subterms in the lower part of the tree of the form $u_1 u_2 \cdots u_k a$, where $u_i \in \{\mathsf{src}, \mathsf{tgt}, \mathsf{id}\}$ and a is a primitive symbol, with all composition operators $\circ_k$ above them.

Running example. By (R2), $\mathsf{tgt}^2(u \circ_1 v) = \mathsf{tgt}^2\, u$, so after Step 1 we obtain $(\mathsf{id}^2\, \mathsf{tgt}^2\, u) \circ_1 u$.

Step 2: normalize unary leaves. On each maximal unary leaf, we exhaustively apply (R6)–(R9). We apply Rules (R6) and (R7) to eliminate any occurrence of id below an occurrence of src or tgt and (R8) and (R9) to change any mixed composition src and tgt to a composition of either src or tgt alone; for example, $\mathsf{src}\, \mathsf{tgt}\, \mathsf{src}\, \mathsf{src}\, \mathsf{tgt}\, a$ becomes $\mathsf{src}^5\, a$. When this step is completed, all subterms involving only the unary operators (i.e. all unary leaves) are in normal form $\mathsf{id}^k\, \mathsf{end}^m a$.

Running example. By (R9), $\mathsf{tgt}\, \mathsf{src}\, a = \mathsf{tgt}^2\, a$, hence $v = u$. Also $\mathsf{tgt}^2\, u = \mathsf{tgt}^2\, a$ by (R7), so $\mathsf{id}^2\, \mathsf{tgt}^2\, u = u$ and the term becomes $u \circ_1 u$.

Step 3: eliminate compositions bottom up. We now traverse the composition tree bottom up to eliminate compositions. Whenever we reach a lowest composition node

$$u \circ_k v$$

whose children are normal forms $u = \mathrm{id}^\ell \, \mathrm{end}_1^m a$ and $v = \mathrm{id}^i \, \mathrm{end}_2^j b$, we test whether the composition is typeable in the free model by checking the boundary condition

$$\mathrm{src}^k \, u = \mathrm{tgt}^k \, v.$$

Using the normal form representation, this boundary check reduces to a constant number of comparisons on the integers ℓ, m, i, j, k and the symbols a, b (details in Lemma 4.4 below). If the boundary check fails, we halt and declare failure. If it succeeds, then in the free model the composite is equal to one of the two operands, and we replace the node by that operand. This strictly decreases the number of composition operators, so Step 3 terminates.

Running example. The boundary check for $u \circ_1 u$ is $\mathrm{src}^2 \, u = \mathrm{tgt}^2 \, u$, which holds since both sides reduce to $\mathrm{tgt}^2 \, a$. By Lemma 4.4(a), $u \circ_1 u = u$.

Lemma 4.4 (Bottom-up composition on normal forms). *Let $u = \mathrm{id}^\ell \, \mathrm{end}_1^m a$ and $v = \mathrm{id}^i \, \mathrm{end}_2^j b$ be normal forms and let $\dim u = \dim v = n \geq k$. In the free model, the composition $u \circ_{n-k} v$ is typeable iff $a = b$ and one of the following holds:*

(a) *$k \leq \ell$ and $k \leq i$ and $u = v$ (equivalently, the conditions of Lemma 4.2 hold); in this case $u \circ_{n-k} v = u$.*
(b) *$k \leq \ell$ and $i < k$ and $\ell = k$ and $\mathrm{end}_1 = \mathrm{tgt}$ and $m = j + k - i$; in this case $u \circ_{n-k} v = u$.*
(c) *$\ell < k$ and $k \leq i$ and $i = k$ and $\mathrm{end}_2 = \mathrm{src}$ and $j = m + k - \ell$; in this case $u \circ_{n-k} v = v$.*

If $\ell < k$ and $i < k$ then the composition is not typeable in the free model.

Theorem 4.5. *For every well formed term e, each of Steps 1–3 either fails, in which case e is undefined in the free model, or returns a normal form $\mathrm{NF}(e)$ such that $e = \mathrm{NF}(e)$ in the free model. Moreover, for any well formed e_1, e_2 of the same dimension,*

$$e_1 = e_2 \text{ in the free model}$$
$$\Leftrightarrow \mathrm{NF}(e_1) \text{ and } \mathrm{NF}(e_2) \text{ both succeed and are equal.}$$

Proof sketch. Steps 1 and 2 are rewriting using oriented axioms, hence preserve equality in the free model. Step 3 eliminates compositions bottom up. Each elimination step is justified by Lemma 4.4 together with the unit laws (R10)–(R11), and strictly decreases the number of composition operators, hence terminates. For completeness, any derivation in the free model can be replayed as rewriting to normal form because all rules are oriented and Step 3 eliminates every typeable composition. Finally, equality of returned normal forms is decided by Lemma 4.2. □

To conclude, we have shown

Theorem 4.6. *EqCat_∞ is decidable.*

5 UNDECIDABILITY OF GLOBAL VALIDITY

In this section we show that equality of meaningful terms is undecidable in general.

Let X be a set of variables, each with an assigned dimension $\dim x \in \mathbb{N}$. Recall that an *interpretation* in an ∞-category C is specified by a map $\sigma : X \to C$ respecting dimension. Let $\mathcal{EQ}$ denote the set of well-formed equations $e_1 = e_2$ over X such that for every ∞-category C and for every interpretation $\sigma : X \to C$, either one of e_1 or e_2 is undefined or both are defined and are equal.

Theorem 5.1. *Membership in $\mathcal{EQ}$ is undecidable.*

Proof. The proof is by reduction from the triviality problem for monoid presentations, a known undecidable problem [18]. Given a finite monoid presentation consisting of generators and relations, we show how to construct an expression that is in $\mathcal{EQ}$ iff the given presentation represents the trivial monoid.

Let Σ be a collection of variables of dimension 1 representing 1-cells. The equations

$$s \circ_0 t = s \circ_0 t, \ s, t \in \Sigma$$

will all be well typed and valid under an interpretation σ iff the sources and targets of all the $\sigma(s)$ are the same 0-cell. Thus the $\sigma(s)$ for $s \in \Sigma$ generate a monoid. We implicitly include these equations in the construction below.

A word $w = s_1 \cdots s_n \in \Sigma^*$ is represented by the composition $s_1 \circ_0 \cdots \circ_0 s_n$. For $w_1, w_2 \in \Sigma^*$, the monoid relation $w_1 = w_2$ is represented by the composition $\mathrm{id}\, w_1 \circ_1 \mathrm{id}\, w_2$. This is a composition of 2-cells. It will be defined under the interpretation σ iff $\sigma(\mathrm{src}\,\mathrm{id}\, w_1) = \sigma(\mathrm{tgt}\,\mathrm{id}\, w_2)$, that is, $\sigma(w_1) = \sigma(w_2)$.

To represent multiple monoid relations, we compose via $\circ_0$, so that the set

$$\{w_1^1 = w_2^1, \ w_1^2 = w_1^2, \ \ldots, \ w_1^n = w_2^n\}$$

is represented by the composition

$$(\mathrm{id}\, w_1^1 \circ_1 \mathrm{id}\, w_2^1) \circ_0 (\mathrm{id}\, w_1^2 \circ_1 \mathrm{id}\, w_2^2) \circ_0 \cdots \circ_0 (\mathrm{id}\, w_1^n \circ_1 \mathrm{id}\, w_2^n).$$

$$\begin{tikzcd}[column sep=large]
\bullet \arrow[r, bend left=60, "w_2^1"{name=A}] \arrow[r, ""{name=M}] \arrow[r, bend right=60, "w_1^1"'{name=B}] \arrow[from=A, to=M, phantom, "\Downarrow \mathrm{id}\, w_2^1"] \arrow[from=M, to=B, phantom, "\Downarrow \mathrm{id}\, w_1^1"] & \bullet \arrow[r, bend left=60, "w_2^2"{name=C}] \arrow[r, ""{name=N}] \arrow[r, bend right=60, "w_1^2"'{name=D}] \arrow[from=C, to=N, phantom, "\Downarrow \mathrm{id}\, w_2^2"] \arrow[from=N, to=D, phantom, "\Downarrow \mathrm{id}\, w_1^2"] & \bullet & \cdots & \bullet \arrow[r, bend left=60, "w_2^n"{name=E}] \arrow[r, ""{name=P}] \arrow[r, bend right=60, "w_1^n"'{name=F}] \arrow[from=E, to=P, phantom, "\Downarrow \mathrm{id}\, w_2^n"] \arrow[from=P, to=F, phantom, "\Downarrow \mathrm{id}\, w_1^n"] & \bullet
\end{tikzcd}$$

A presentation of the trivial monoid with generators $\Sigma = \{s_1, \ldots, s_n\}$ is of course

$$(\mathrm{id}\, s_1 \circ_1 \mathrm{id}\,\mathrm{id}\,\mathrm{src}\, s_1) \circ_0 \cdots \circ_0 (\mathrm{id}\, s_n \circ_1 \mathrm{id}\,\mathrm{id}\,\mathrm{src}\, s_n).$$

Fixing an arbitrary monoid presentation

$$\{w_1^1 = w_2^1, \ w_1^2 = w_1^2, \ \ldots, \ w_1^n = w_2^n\},$$

if it is a presentation of the trivial monoid, then the equation

$$(\mathrm{id}\, w_1^1 \circ_1 \mathrm{id}\, w_2^1) \circ_0 (\mathrm{id}\, w_1^2 \circ_1 \mathrm{id}\, w_2^2) \circ_0 \cdots \circ_0 (\mathrm{id}\, w_1^n \circ_1 \mathrm{id}\, w_2^n)$$
$$= (\mathrm{id}\, s_1 \circ_1 \mathrm{id}\,\mathrm{id}\,\mathrm{src}\, s_1) \circ_0 \cdots \circ_0 (\mathrm{id}\, s_m \circ_1 \mathrm{id}\,\mathrm{id}\,\mathrm{src}\, s_n)$$

holds in any ∞-category C, provided the elements of Σ are assigned to elements of C in such a way that both sides of the equation are well typed. Conversely, if there is an ∞-category C and interpretation of the elements of Σ by elements of C such that the interpreted equation does not hold in C, then because all 1-cells in Σ are composable with each other, this witnesses that the given presentation is not a presentation of the trivial monoid. Therefore the problem of deciding whether a finite monoid presentation represents the trivial monoid reduces to the membership problem for $\mathcal{EQ}$, and consequently $\mathcal{EQ}$ is undecidable. □

Theorem 5.2. *The problem with input an expression e in the untyped language of ∞-categories and output the most general assignment of types guaranteeing that e is well-typed is undecidable.*

Proof. Essentially the same reduction as in Theorem 5.1. The expression constructed in that theorem to encode a monoid presentation has $1_{1_*} \Rightarrow 1_{1_*}$ as its most general type precisely when the presented monoid is trivial. □

The results of §4 and §5 may appear contradictory, but they are not. The key difference is that the definition of $\mathcal{EQ}$ says "*If* e_1 and e_2 are defined in C,..." which essentially allows the encoding of a Horn formula involving monoid equations, whereas §4 deals exclusively with the equational theory of the free model.

6 ADEQUACY OF STRICT ω-CATEGORIES

Models of EqCat_∞ coincide with the notion of strict ω-category (see e.g., [6]). Strict ω-categories are the fully strict form of higher categories: all compositions, units, and interchange laws hold on the nose rather than only up to higher cells. Although strictness is often too rigid for directly modeling weak or homotopical phenomena, strict ω-categories are widely used as algebraic and combinatorial objects, for example via polygraphs and higher dimensional rewriting [? ?]. They serve as a convenient setting for higher categorical rewriting and coherence computations through polygraphs [?], and they connect to simplicial and combinatorial formalisms via nerve constructions, where strict ω-categories correspond to structured simplicial sets such as complicial sets [?]. Finally, strict globular ω-categories have been used in semantics of concurrency, where they model execution paths together with higher homotopies between them [?].

Concretely, a strict ω-category (internal to Set) is a globular set $C \triangleq (\cdots \rightrightarrows C_3 \rightrightarrows C_2 \rightrightarrows C_1 \rightrightarrows C_0)$ where for all $k > \ell > m$: $(C_k \rightrightarrows C_\ell)$ is a category and $(C_k \rightrightarrows C_\ell \rightrightarrows C_m)$ is a strict 2-category. The strong reading of our typed equational theory interprets equations as definitional equality, and the adequacy theorem below shows that this strong semantics is exactly the classical strict ω-categorical one.

Theorem 6.1. *An* EqCat_∞ *model is exactly a strict ω-category.*

Proof sketch. ($\Rightarrow$) Assume a EqCat_∞ model is a strict ω-category. The axioms of an EqCat_∞ model are designed to directly encode the necessary structures:

The fundamental source/target identities (IC1, IC2) establish the globular set property. Composition and identity axioms (e.g., IC3, IC4, IC9, IC11, IC12) for different dimension levels directly form the category structures $(C_k \rightrightarrows C_\ell)$. The functoriality of source/target (IC7, IC8),the exchange law (IC10), and the identity law (IC13) are precisely the properties required for the strict 2-category structures $(C_k \rightrightarrows C_\ell \rightrightarrows C_m)$.

($\Leftarrow$) Conversely, if we begin with the definition of a strict ω-category (a globular set where appropriate hierarchical levels form strict categories and 2-categories), all the EqCat_∞ axioms are immediate consequences. Each EqCat_∞ axiom directly corresponds to a property guaranteed by the globular, strict category, or strict 2-category structure at the relevant dimension levels. The EqCat_∞ axioms simply make explicit these inherent properties.

□

7 HOMOTOPY MODELS

There are many approaches to weak higher-order categories, e.g. [1, 8, 19]. In this section we present a weak model of our axioms (for the index set $\mathbb{N}$ only) in which cells are boundary-preserving homotopies and equivalences between cells are witnessed by higher-order cells. Our goal is not to propose a new definition of weak higher categories. Rather, we show that the typed equational calculus EqCat_∞ admits a concrete homotopical reading. In this reading, equalities are interpreted by explicit witnesses, namely higher cells, and the only semantic structure we assume is a small collection of closure operations on homotopies that suffices to interpret the operations and axioms of the calculus.

Let X be a topological space. Let I denote the closed real unit interval $[0, 1]$. A *homotopy model* is an ∞-category-like structure whose n-cells are certain continuous maps $I^n \to X$ in which equality is interpreted as a certain equivalence to be defined precisely below. We will show that the axioms of §3 are sound for these interpretations.

7.1 Homotopies

An *n-dimensional homotopy* is a continuous map $\alpha : I^n \to X$. Not all homotopies are n-cells, but only those satisfying a certain boundary-preservation condition. We define $\dim \alpha = n$. The set of n-dimensional homotopies is denoted H_n. Thus a 0-dimensional homotopy $a : I^0 \to X$ is simply a point of X, and we write $a \in X$. A 1-dimensional homotopy $f : I \to X$ is a continuous path in X from $f(0)$ to $f(1)$. A 2-dimensional homotopy is a continuous map $\alpha : I^2 \to X$. Viewing its first argument as time, the map α can be regarded as a continuous deformation of the path $\lambda x \,.\, \alpha(0, x)$ to $\lambda x \,.\, \alpha(1, x)$ with intermediate states $\lambda x \,.\, \alpha(t, x)$.

Given an n-dimensional homotopy $\alpha : I^n \to X$ and $1 \le k \le n$, let $\alpha^{(k)}$ be α curried on its k-th component:

$$\alpha^{(k)} : I \to I^{n-1} \to X$$
$$\alpha^{(k)}(x_k)(x_1, \ldots, x_{k-1}, x_{k+1}, \ldots, x_n) = \alpha(x_1, \ldots, x_n).$$

We always think of the first component as time and often write t for x_1, so

$$\alpha^{(1)}(t)(x_2, \ldots, x_n) = \alpha(t, x_2, \ldots, x_n).$$

For $\alpha \in H_n$, $x \in I$, and $1 \le k \le n$, $\alpha^{(k)}(x)$ is called a *section* of α. A *face* of α is a section $\alpha^{(k)}(b)$ with $b \in \{0, 1\}$. Two special faces are $\operatorname{src} \alpha = \alpha^{(1)}(0)$ and $\operatorname{tgt} \alpha = \alpha^{(1)}(1)$, called the *source* and *target* of α, respectively. We thus view α as a continuous deformation of $\operatorname{src} \alpha$ to $\operatorname{tgt} \alpha$, passing through intermediate states $\alpha^{(1)}(t) : I^{n-1} \to X$. For $n \ge 2$, the maps $\alpha^{(2)}(0)$ and $\alpha^{(2)}(1)$ describe the trajectory of $\operatorname{src} \operatorname{src} \alpha$ and $\operatorname{tgt} \operatorname{src} \alpha$ throughout the deformation.

Lower-dimensional sections and faces can be defined inductively. For $1 \le i_1 < i_2 < \cdots < i_k \le n$ and $\alpha \in H_n$, let $\alpha^{(i_1,\ldots,i_k)} : I^k \to I^{n-k} \to X$ be α curried on components $i_1, \ldots, i_k$:

$$\alpha^{(i_1,\ldots,i_k)}(x_1, \ldots, x_k) = \alpha^{(i_k)}(x_k)^{(i_1,\ldots,i_{k-1})}(x_1, \ldots, x_{k-1}).$$

The function $\alpha^{(i_1,\ldots,i_k)}(x_1, \ldots, x_k) : I^{n-k} \to X$ is an $n-k$-dimensional section and represents α partially evaluated at values $x_1, \ldots, x_k$ for the variables at positions $i_1, \dots, i_k$, respectively. For example,

$$\alpha^{(1,3)}(a,b)(x,y,z) = \alpha(a,x,b,y,z).$$

An $\alpha \in H_n$ is completely determined by its sections $\alpha^{(i_1,\dots,i_k)}(x_1,\dots,x_k)$ for any fixed selection of indices $1 \le i_1 < \cdots < i_k \le n$. One can define α by specifying these sections and verifying that the corresponding uncurried version is continuous. An essential tool in this regard is Lemma E.1 in the appendix.

A note on notation: When using prefix and postfix unary operators together, the postfix takes precedence. For example, $\operatorname{src}\alpha^{(1)}(t)$ should always be read as $\operatorname{src}(\alpha^{(1)}(t))$. To obtain $(\operatorname{src}\alpha)^{(1)}(t)$, parentheses must be used. As before, binary operators have least precedence; thus $\operatorname{src}\alpha \circ_k \operatorname{tgt}\alpha$ should be read as $(\operatorname{src}\alpha) \circ_k (\operatorname{tgt}\alpha)$.

7.2 Identities and Composition

If $\alpha \in H_n$, define $\operatorname{id}\alpha$ by $(\operatorname{id}\alpha)^{(1)}(t) = \alpha$; that is, $(\operatorname{id}\alpha)^{(1)}$ is the constant function $\lambda t.\alpha$. Then $\operatorname{id}\alpha \in H_{n+1}$, as it is the composition of $\alpha : I^n \to X$ with the continuous projection $\lambda t, \overline{x}.\overline{x} : I^{n+1} \to I^n$. An *identity* is any homotopy of the form $\operatorname{id}\alpha$; that is, β is an identity iff the value of $\beta^{(1)}(t)$ is independent of t. Note that β is an identity iff $\beta = \operatorname{id}\operatorname{src}\beta$, and also iff $\beta = \operatorname{id}\operatorname{tgt}\beta$.

Both src and tgt are left inverses of id:

$$\operatorname{src}\operatorname{id}\alpha = (\operatorname{id}\alpha)^{(1)}(0) = \alpha \qquad \operatorname{tgt}\operatorname{id}\alpha = (\operatorname{id}\alpha)^{(1)}(1) = \alpha. \tag{4}$$

If $\alpha, \beta \in H_n$ and $\operatorname{tgt}\alpha = \operatorname{src}\beta$, we can compose them as follows:

$$(\beta \circ_{n-1} \alpha)^{(1)}(t) = \begin{cases} \alpha^{(1)}(2t), & \text{if } 0 \le t \le \frac{1}{2} \\ \beta^{(1)}(2t-1), & \text{if } \frac{1}{2} \le t \le 1. \end{cases}$$

There is no discontinuity at $t = 1/2$, because $\alpha^{(1)}(1) = \operatorname{tgt}\alpha = \operatorname{src}\beta = \beta^{(1)}(0)$. The source and target of $\beta \circ_{n-1} \alpha$ are

$$\begin{aligned} \operatorname{src}(\beta \circ_{n-1} \alpha) &= (\beta \circ_{n-1} \alpha)^{(1)}(0) = \alpha^{(1)}(0) = \operatorname{src}\alpha \\ \operatorname{tgt}(\beta \circ_{n-1} \alpha) &= (\beta \circ_{n-1} \alpha)^{(1)}(1) = \beta^{(1)}(1) = \operatorname{tgt}\beta. \end{aligned} \tag{5}$$

More generally, if $\alpha^{(k)}(1) = \beta^{(k)}(0)$, we can compose them along their k-th $n-1$-dimensional faces:

$$(\beta \circ_{n-k} \alpha)^{(k)}(x) = \begin{cases} \alpha^{(k)}(2x), & \text{if } 0 \le x \le 1/2 \\ \beta^{(k)}(2x-1), & \text{if } 1/2 \le x \le 1. \end{cases} \tag{6}$$

For $k > 1$,

$$\begin{aligned} (\operatorname{src}\alpha)^{(k-1)}(1) &= \alpha^{(1)}(0)^{(k-1)}(1) = \alpha^{(k)}(1)^{(1)}(0) \\ &= \beta^{(k)}(0)^{(1)}(0) = \beta^{(1)}(0)^{(k-1)}(0) \\ &= (\operatorname{src}\beta)^{(k-1)}(0), \end{aligned}$$

so $\operatorname{src}\beta \circ_{n-k} \operatorname{src}\alpha$ exists, and

$$\begin{aligned} \operatorname{src}(\beta \circ_{n-k} \alpha) &= (\beta \circ_{n-k} \alpha)^{(1)}(0) \qquad (7) \\ &= \begin{cases} \alpha^{(k)}(2x)^{(1)}(0), & \text{if } 0 \le x \le 1/2 \\ \beta^{(k)}(2x-1)^{(1)}(0), & \text{if } 1/2 \le x \le 1 \end{cases} \\ &= \begin{cases} \alpha^{(1)}(0)^{(k-1)}(2x), & \text{if } 0 \le x \le 1/2 \\ \beta^{(1)}(0)^{(k-1)}(2x-1), & \text{if } 1/2 \le x \le 1 \end{cases} \\ &= \begin{cases} (\operatorname{src}\alpha)^{(k-1)}(2x), & \text{if } 0 \le x \le 1/2 \\ (\operatorname{src}\beta)^{(k-1)}(2x-1), & \text{if } 1/2 \le x \le 1 \end{cases} \\ &= \operatorname{src}\beta \circ_{n-k} \operatorname{src}\alpha. \qquad (8) \end{aligned}$$

Similarly, $\operatorname{tgt}(\beta \circ_{n-k} \alpha) = \operatorname{tgt}\beta \circ_{n-k} \operatorname{tgt}\alpha$. It follows inductively that

$$\operatorname{src}^k(\beta \circ_{n-k} \alpha) = \operatorname{src}^k \alpha \qquad \operatorname{tgt}^k(\beta \circ_{n-k} \alpha) = \operatorname{tgt}^k \beta. \tag{9}$$

Also, for n-cells α, β and $1 \le k \le n$,

$$\begin{aligned} &(\operatorname{id}\beta \circ_{n-k} \operatorname{id}\alpha)^{(1)}(t)^{(k)}(s) \\ &= (\operatorname{id}\beta \circ_{n-k} \operatorname{id}\alpha)^{(k+1)}(s)^{(1)}(t) \\ &= \begin{cases} (\operatorname{id}\alpha)^{(k+1)}(2s)^{(1)}(t) & \text{if } 0 \le s \le 1/2 \\ (\operatorname{id}\beta)^{(k+1)}(2s-1)^{(1)}(t) & \text{if } 1/2 \le s \le 1 \end{cases} \\ &= \begin{cases} (\operatorname{id}\alpha)^{(1)}(t)^{(k)}(2s) & \text{if } 0 \le s \le 1/2 \\ (\operatorname{id}\beta)^{(1)}(t)^{(k)}(2s-1) & \text{if } 1/2 \le s \le 1 \end{cases} \\ &= \begin{cases} \alpha^{(k)}(2s) & \text{if } 0 \le s \le 1/2 \\ \beta^{(k)}(2s-1) & \text{if } 1/2 \le s \le 1 \end{cases} \\ &= (\beta \circ_{n-k} \alpha)^{(k)}(s), \end{aligned}$$

thus $(\operatorname{id}\beta \circ_{n-k} \operatorname{id}\alpha)^{(1)}(t) = \beta \circ_{n-k} \alpha$. As this is independent of t,

$$\operatorname{id}(\beta \circ_{n-k} \alpha) = \operatorname{id}\beta \circ_{n-k} \operatorname{id}\alpha. \tag{10}$$

Viewing $\alpha \in H_n$ for $n \ge 1$ as a deformation from $\operatorname{src}\alpha$ to $\operatorname{tgt}\alpha$, there is an $\alpha^{\leftarrow} \in H_n$ giving a deformation in the reverse direction, namely $(\alpha^{\leftarrow})^{(1)}(t) = \alpha^{(1)}(1-t)$. This is an involution in the sense that $(\alpha^{\leftarrow})^{\leftarrow} = \alpha$.

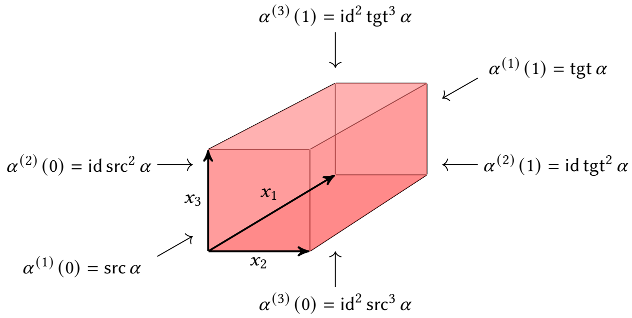
7.3 Cells

A *cell* is a homotopy satisfying a certain boundary-preservation condition. All 0- and 1-dimensional homotopies are cells by definition. For $n \ge 2$, an n-*cell* is an $\alpha \in H_n$ such that

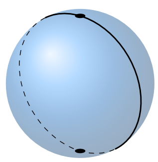
- for all $t \in I$, $\alpha^{(1)}(t)$ is (inductively) an $n-1$-cell;
- for $b \in \{0,1\}$, $\alpha^{(2)}(b)$ is an identity.

The latter condition says that $\alpha^{(2)}(b)^{(1)}(t)$ is independent of t. It is equivalent to the condition $\alpha^{(2)}(b) = \operatorname{id}\operatorname{src}\alpha^{(2)}(b)$. For $b = 0$, it is equivalent to the condition $\alpha^{(2)}(0) = \operatorname{id}\operatorname{src}^2\alpha$, since $\operatorname{src}\alpha^{(2)}(0) = \alpha^{(1)}(0)^{(2)}(0) = \alpha^{(1)}(0)^{(1)}(0) = \operatorname{src}^2\alpha$.

One thinks of a cell as an open set whose boundary comprises lower-dimensional cells. Fig. 4 illustrates a cell $\alpha : I^3 \to X$ of dimension 3. Fig. 4(a) is the rectilinear diagram that represents I^3. Each point of this set is mapped to a point of the ambient space X. The source and target of α are maps $I^2 \to X$ and are the front and back faces as indicated. The vertical line labeled x_3 is $\operatorname{src}^2\alpha$. Each vertical line along the left face maps to the same points of



(a) Rectilinear view.



(b) Spherical view.

Figure 4: A 3-cell $\alpha : I^3 \to X$ in rectilinear and spherical representations.

X as this one, including the leftmost line of the back face, which is $\mathrm{src}\,\mathrm{tgt}\,\alpha$. Thus, although this face is 2-dimensional, it actually represents a replicated one-dimensional line in X. Similarly, every point of the top face maps to the same point of X, that is, α is constant on the top face, representing a single point of X. The rectilinear representation of cells allows for a convenient calculus for formalizing compositions.

Lemma 7.1. *Let $\alpha \in H_n$. The following are equivalent:*

(i) *α is a cell;*
(ii) *for all $2 \le k \le n$, for all $x_1, \ldots, x_{k-1} \in I^{k-1}$ and $b \in \{0, 1\}$,*
$$\alpha^{(1,2,\ldots,k)}(x_1, \ldots, x_{k-1}, b) = \alpha^{(1,2,\ldots,k)}(b, \ldots, b, b); \tag{11}$$
(iii) *for all $1 \le k \le n$, $\alpha^{(k)}(0) = \mathrm{id}^{k-1}\,\mathrm{src}^k\,\alpha$ and $\alpha^{(k)}(1) = \mathrm{id}^{k-1}\,\mathrm{tgt}^k\,\alpha$;*
(iv) *for all $1 \le m \le n-1$ and $b \in \{0, 1\}$, the value of $\alpha^{(m,m+1)}(x, b)$ is independent of x.*

It follows from the characterization of Lemma 7.1(iii) that
$$\begin{aligned}\mathrm{src}\,\mathrm{tgt}\,\alpha &= \alpha^{(1)}(1)^{(1)}(0) = \alpha^{(2)}(0)^{(1)}(1) = \mathrm{tgt}\,\mathrm{id}\,\mathrm{src}^2\,\alpha = \mathrm{src}^2\,\alpha\\ \mathrm{tgt}\,\mathrm{src}\,\alpha &= \alpha^{(1)}(0)^{(1)}(1) = \alpha^{(2)}(1)^{(1)}(0) = \mathrm{src}\,\mathrm{id}\,\mathrm{tgt}^2\,\alpha = \mathrm{tgt}^2\,\alpha.\end{aligned} \tag{12}$$

Lemma 7.2. *All sections of cells are cells.*

7.4 Reparameterization

Reparameterization is the main source of coherence witnesses in our semantics. It turns a cell α into a higher cell by varying α along a chosen coordinate according to an explicit continuous map $f : I^2 \to I$. In the soundness proofs, a small fixed family of such maps is used to witness associativity, identities, inverses, and exchange up to $\simeq_1$.

Definition 7.3. Let $\alpha \in C_n$, $1 \le k \le n$. Let $f : I^2 \to I$ be a continuous function such that for $b \in \{0, 1\}$, $f^{(2)}(b)$ is the constant function $\mathrm{id}\,c$ for some $c \in \{0, 1\}$. Define $\mathrm{rpm}(\alpha, f, k) \in H_{n+1}$ by
$$\mathrm{rpm}(\alpha, f, k)^{(1,k+1)}(t, s) = \alpha^{(k)}(f(t, s)),$$
that is,
$$\mathrm{rpm}(\alpha, f, k)^{(1,k+1)} = \alpha^{(k)} \circ f : I^2 \to I^{n-1} \to X.$$
This operation is called *reparameterization.*

The continuity of $\mathrm{rpm}(\alpha, f, k)$ follows from Lemma E.1, because $\alpha^{(k)}$ and hence the composition $\alpha^{(k)} \circ f$ are continuous with respect to the compact-open topology in $I^{n-1} \to X$. Thus for any n-cell α and suitable f as defined in Definition 7.3, $\mathrm{rpm}(\alpha, f, k) \in H_{n+1}$, but we will prove below (Lemma 7.5) that it is in C_{n+1}.

Lemma 7.4. *For $\alpha \in C_n$, $f : I^2 \to I$, $x, y \in I$, and $1 \le k \le n$,*
$$\begin{aligned}&\mathrm{rpm}(\alpha, f, k)^{(m+1,m+2)}(x, y)\\ &= \begin{cases}\mathrm{rpm}(\alpha^{(m,m+1)}(x, y), f, k-2) & \text{if } 1 \le m \le k-2\\ \mathrm{rpm}(\alpha^{(m,m+1)}(x, y), f, k) & \text{if } k+1 \le m \le n-1.\end{cases}\end{aligned}$$
For the remaining two values of m not covered above,
$$\begin{aligned}\mathrm{rpm}(\alpha, f, k)^{(k,k+1)}(x, s)^{(1)}(t) &= \alpha^{(k-1,k)}(x, f(t, s))\\ \mathrm{rpm}(\alpha, f, k)^{(k+1,k+2)}(s, y)^{(1)}(t) &= \alpha^{(k,k+1)}(f(t, s), y).\end{aligned}$$

Lemma 7.5. *Let $f : I^2 \to I$ be a continuous function such that for $b \in \{0, 1\}$, $f^{(2)}(b)$ is a constant function $\mathrm{id}\,c$ for some $c \in \{0, 1\}$. Then for all $\alpha \in C_n$ and $1 \le k \le n$, $\mathrm{rpm}(\alpha, f, k) \in C_{n+1}$.*

7.5 Definition of Homotopy Models

Let X be a topological space. A *homotopy model* C is an ∞-category-like structure consisting of a specified base set of n-cells $\alpha : I^n \to X$ for each n and satisfying certain minimal closure conditions needed for elementary constructions. The set of n-cells of the model C is denoted C_n. Specifically,

- If $\alpha \in C_n$ for $n \ge 1$, then $\mathrm{src}\,\alpha$ and $\mathrm{tgt}\,\alpha$ are in C_{n-1}.
- If $\alpha, \beta \in C_n$, $1 \le k \le n$, and $\alpha^{(k)}(1) = \beta^{(k)}(0)$, then $\beta \circ_{n-k} \alpha \in C_n$.
- If $1 \le k \le n+1$, $\alpha \in C_n$, and $f : I^2 \to I$ is one of a small collection of continuous functions listed below, then the reparameterizations $\mathrm{rpm}(\alpha, f, k)$ and $\mathrm{rpm}(\alpha, f, k)^{\leftarrow}$ are in C_{n+1}.

Most of the reparameterizations we will need are of the form $f_{ab}(t, s) = t\,c_{ab}(s) + (1-t)s$, where
$$c_{ab}(s) = \begin{cases} sa/b, & \text{if } 0 \le s \le b\\ a + (s-b)(1-a)/(1-b), & \text{if } b \le s \le 1\end{cases} \tag{13}$$
for $0 \le a \le 1$ and $0 < b < 1$. The function c_{ab} is continuous at b, as both branches give the value a. Some special cases are
$$\begin{aligned} c_{1/2,1/2}(x) &= x\\ c_{0,1/2}(x) &= \begin{cases}0, & \text{if } 0 \le x \le 1/2\\ 2x-1, & \text{if } 1/2 \le x \le 1\end{cases}\\ c_{1,1/2}(x) &= \begin{cases}2x, & \text{if } 0 \le x \le 1/2\\ 1, & \text{if } 1/2 \le x \le 1\end{cases}\end{aligned}$$

$$c_{1/2,1/3}(x) = \begin{cases} 3x/2, & \text{if } 0 \le x \le 1/3 \\ (3x+1)/4, & \text{if } 1/3 \le x \le 1 \end{cases}$$

$$c_{1/2,2/3}(x) = \begin{cases} 3x/4, & \text{if } 0 \le x \le 2/3 \\ (3x-1)/2, & \text{if } 2/3 \le x \le 1 \end{cases}$$

and in fact, these are all the reparameterization of this kind we will need. Note that $c_{ab}(b) = b$ and $f_{ab}(t,b) = b$, so Lemma 7.5 applies. These will be used for associativity of $\circ_k$ and identities.

We will need one more reparameterization to deal with inverses, namely

$$u(t,s) = \begin{cases} 2ts, & \text{if } 0 \le s \le 1/2 \\ 2t(1-s), & \text{if } 1/2 \le s \le 1. \end{cases} \tag{14}$$

Note that $u(t,0) = u(t,1) = 0$, so Lemma 7.5 applies.

Our closure rules above did not contain the rule $\alpha \in C_n \Rightarrow \mathrm{id}\,\alpha \in C_{n+1}$, but we get this from $\mathrm{rpm}(\alpha, f_{1/2,1/2}, 1)$:

$$\begin{aligned} \mathrm{rpm}(\alpha, f_{1/2,1/2}, 1)^{(1)}(t)^{(1)}(x) &= \mathrm{rpm}(\alpha, f_{1/2,1/2}, 1)^{(1,2)}(t,x) \\ &= \alpha^{(1)}(tx + (1-t)x) = \alpha^{(1)}(x), \end{aligned}$$

so $\mathrm{rpm}(\alpha, f_{1/2,1/2}, 1)^{(1)}(t) = \alpha$, so $\mathrm{rpm}(\alpha, f_{1/2,1/2}, 1) = \mathrm{id}\,\alpha$.

7.6 Interpretation of Equality

Equality in a homotopy model is not interpreted as identity, but as a collection of relations $\simeq_k$. We are most interested in $\simeq_1$. Accordingly, we treat an equation judgment as a request for a witness one dimension higher. The relations $\simeq_k$ make this explicit: $\alpha \simeq_k \beta$ holds precisely when there exists a suitable $(n+1)$-cell φ connecting α to β, as formalized below.

Definition 7.6. Let us call a cell $\alpha \in C_n$ *invertible at level* k if $\alpha^{\leftarrow k} \in C_n$, where $\alpha^{(\leftarrow k)}(s) = \alpha^{(k)}(1-s)$. (Here $\alpha^{(\leftarrow k)}(s)$ is shorthand for $(\alpha^{\leftarrow k})^{(k)}(s)$.) For $1 \le k \le n+1$, define

$$\alpha \simeq_k \beta \iff \begin{array}{l} \text{there exists } \varphi \in C_{n+1} \text{ invertible at level } k \text{ such that} \\ \alpha = \varphi^{(k)}(0) \text{ and } \beta = \varphi^{(k)}(1). \end{array}$$

Equivalently by Lemma 7.1,

$$\alpha \simeq_k \beta \iff \begin{array}{l} \text{there exists } \varphi \in C_{n+1} \text{ invertible at level } k \text{ such that} \\ \alpha = \mathrm{id}^{k-1}\,\mathrm{src}^k\,\varphi \text{ and } \beta = \mathrm{id}^{k-1}\,\mathrm{tgt}^k\,\varphi. \end{array}$$

When this occurs, we say that α and β are *equivalent at level* k and that φ is a *witness* for the equivalence.

For $1 \le k \le n$, let C_n^k denote those cells $\alpha \in C_n$ such that $\alpha = \mathrm{id}^{k-1}\,\mathrm{src}^{k-1}\,\alpha$; equivalently, such that $\alpha = \mathrm{id}^{k-1}\,\mathrm{tgt}^{k-1}\,\alpha$. If $\alpha = \varphi^{(k)}(0) = \mathrm{id}^{k-1}\,\mathrm{src}^k\,\varphi$ or if $\alpha = \varphi^{(k)}(1) = \mathrm{id}^{k-1}\,\mathrm{tgt}^k\,\varphi$, then α is in C_n^k, so $\simeq_k$ only holds between cells in C_n^k. Note that $C_n^1 = C_n$ and $C_n^{k+1} \subseteq C_n^k$.

Lemma 7.7. *The relation $\simeq_k$ is an equivalence relation on C_n^k and the relations $\simeq_k$ form a stratified congruence with respect to the operations $\mathrm{src}, \mathrm{tgt} : C_n \to C_{n-1}$ for $n \ge 1$, $\mathrm{id} : C_n \to C_{n+1}$, and $\circ_m \colon C_n \times C_n \to C_n$ for $0 \le m \le n-1$ in the following sense: for $\alpha, \beta \in C_n^k$,*

(i) *If $\alpha \simeq_k \beta$ and $k \ge 2$, then $\mathrm{src}\,\alpha \simeq_{k-1} \mathrm{src}\,\beta$ and $\mathrm{tgt}\,\alpha \simeq_{k-1} \mathrm{tgt}\,\beta$.*
(ii) *If $\alpha \simeq_1 \beta$, then $\mathrm{src}\,\alpha = \mathrm{src}\,\beta$ and $\mathrm{tgt}\,\alpha = \mathrm{tgt}\,\beta$.*
(iii) *If $\alpha \simeq_k \beta$, then $\mathrm{id}\,\alpha \simeq_{k+1} \mathrm{id}\,\beta$.*
(iv) *For $m \ge k$, if $\alpha \simeq_k \alpha'$ and $\beta \simeq_k \beta'$, and if $\beta \circ_{n-m} \alpha$ and $\beta' \circ_{n-m} \alpha'$ exist, then $\beta \circ_{n-m} \alpha \simeq_k \beta' \circ_{n-m} \alpha'$.*

We showed in Lemma 7.7 that $\mathrm{id}\,\alpha$ realizes $\alpha \simeq_k \alpha$ for all k. The following lemma says that the relations $\simeq_k$ are disjoint except for this.

Lemma 7.8. *Let $\alpha, \beta \in C_n$ with $\alpha \simeq_k \beta$.*

(i) *If $\alpha, \beta \in C_n^m$ for some $m > k$, then $\alpha = \beta$.*
(ii) *If $\alpha \simeq_m \beta$ for some $m > k$, then $\alpha = \beta$.*

Lemma 7.9. *The map $\mathrm{id} : C_n \to C_{n+1}$, restricted to C_n^k, is a bijection $\mathrm{id} : C_n^k \to C_{n+1}^{k+1}$ with two-sided inverse $\mathrm{src} = \mathrm{tgt} : C_{n+1}^{k+1} \to C_n^k$. For $\alpha, \beta \in C_{n+1}^{k+1}$, $\alpha \simeq_{k+1} \beta$ if and only if $\mathrm{src}\,\alpha \simeq_k \mathrm{src}\,\beta$, and also if and only if $\mathrm{tgt}\,\alpha \simeq_k \mathrm{tgt}\,\beta$.*

It follows from Lemma 7.9 that $\simeq_k$ for $k \ge 2$ is largely superfluous, as for any cells α, β, $\alpha \simeq_k \beta$ if and only if $\mathrm{src}^{k-1}\,\alpha \simeq_1 \mathrm{src}^{k-1}\,\beta$, and also if and only if $\mathrm{tgt}^{k-1}\,\alpha \simeq_1 \mathrm{tgt}^{k-1}\,\beta$.

8 SOUNDNESS OF THE AXIOMS

In this section we verify the soundness of the axioms in homotopy models. Many of the axioms hold with equality and have been previously noted. In each case below we build the witness cell explicitly, using only composition and the reparameterizations introduced above. These include $\mathrm{src}\,\mathrm{id}\,\alpha = \alpha$ and $\mathrm{tgt}\,\mathrm{id}\,\alpha = \alpha$, established in (4); $\mathrm{src}\,\mathrm{tgt}\,\alpha = \mathrm{src}^2\,\alpha$ and $\mathrm{tgt}\,\mathrm{src}\,\alpha = \mathrm{tgt}^2\,\alpha$, established in (12); $\mathrm{id}(\beta \circ_{n-k} \alpha) = \mathrm{id}\,\beta \circ_{n-k} \mathrm{id}\,\alpha$, established in (10); $\mathrm{src}(\beta \circ_{n-1} \alpha) = \mathrm{src}\,\alpha$ and $\mathrm{tgt}(\beta \circ_{n-1} \alpha) = \mathrm{tgt}\,\beta$ and for $k \ge 2$, $\mathrm{src}(\beta \circ_{n-k} \alpha) = \mathrm{src}\,\beta \circ_{n-k} \mathrm{src}\,\alpha$ and $\mathrm{tgt}(\beta \circ_{n-k} \alpha) = \mathrm{tgt}\,\beta \circ_{n-k} \mathrm{tgt}\,\alpha$, established in (5) and (8) respectively. The exchange law also holds with equality, which we prove below in Lemma 8.5.

The remaining axioms do not necessarily hold with equality, but only with $\simeq_1$. One exception is that $\simeq_1$ is not strictly a congruence with respect to identities, but only satisfies the weaker property $\alpha \simeq_1 \beta \Rightarrow \mathrm{id}\,\alpha \simeq_2 \mathrm{id}\,\beta$. This is to be expected in light of Lemma 7.8.

Lemma 8.1 (Associativity). *Let $1 \le k \le n$ and let α, β, γ be n-cells with $\alpha^{(k)}(1) = \beta^{(k)}(0)$ and $\beta^{(k)}(1) = \gamma^{(k)}(0)$. Then*

$$(\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha \simeq_1 \gamma \circ_{n-k} (\beta \circ_{n-k} \alpha).$$

Lemma 8.2 (Identities). *If α is an n-cell and $1 \le k \le n$, then*

$$\alpha \circ_{n-k} \mathrm{id}^k\,\mathrm{src}^k\,\alpha \simeq_1 \alpha \qquad \mathrm{id}^k\,\mathrm{tgt}^k\,\alpha \circ_{n-k} \alpha \simeq_1 \alpha.$$

Lemma 8.3 (Inverse). *For any cell γ, $\gamma^{\leftarrow k} \circ_{n-k} \gamma$ exists and $\gamma^{\leftarrow k} \circ_{n-k} \gamma \simeq_1 \mathrm{id}^k\,\mathrm{src}^k\,\gamma$.*

Remark 8.4. *Note that this does not imply that $\gamma^{\leftarrow k}$ exists in the model. It is possible for $\gamma^{\leftarrow k} \circ_{n-k} \gamma$ to exist without $\gamma^{\leftarrow k}$.*

The exchange law holds without any work at all:

Lemma 8.5 (Exchange). *For all $1 \le k < m \le n$ and for all n-cells $\alpha, \beta, \gamma, \delta$ satisfying the requisite boundary conditions,*

$$(\alpha \circ_{n-k} \beta) \circ_{n-m} (\gamma \circ_{n-k} \delta) = (\alpha \circ_{n-m} \gamma) \circ_{n-k} (\beta \circ_{n-m} \delta),$$

therefore they are $\simeq_1$ equivalent as witnessed by $\mathrm{id}((\alpha \circ_{n-k} \beta) \circ_{n-m} (\gamma \circ_{n-k} \delta))$.

Lemma 8.6 (Congruence of $\circ_{n-k}$). *Let α, β, γ be n-cells. If $\operatorname{tgt}^k \alpha = \operatorname{tgt}^k \beta = \operatorname{src}^k \gamma$ and $\alpha \simeq_1 \beta$, then $\gamma \circ_{n-k} \alpha \simeq_1 \gamma \circ_{n-k} \beta$. Symmetrically, if $\operatorname{src}^k \alpha = \operatorname{src}^k \beta = \operatorname{tgt}^k \gamma$ and $\alpha \simeq_1 \beta$, then $\alpha \circ_{n-k} \gamma \simeq_1 \beta \circ_{n-k} \gamma$.*

Lemma 8.7. *The following are equivalent:*

- *α and $\beta^{\leftarrow k} \circ_{n-k} \alpha$ are in C_n and $\beta^{\leftarrow k} \circ_{n-k} \alpha \simeq_1 \mathrm{id}^k \operatorname{src}^k \alpha$;*
- *α and β are in C_n and $\alpha \simeq_1 \beta$.*

Up to this point, the main structural laws that are only expected to hold up to equivalence have been justified by explicit witness cells. This keeps the weak reading concrete: coherence appears as higher cell data built inside the model, rather than as an external meta level condition.

In homotopy practice, α and β are called equivalent if there are maps $\varphi : \alpha \longrightarrow \beta$ and $\psi : \beta \longrightarrow \alpha$ whose composites are homotopic to the corresponding identities. In our setting this coincides with with $\simeq$ interpreted as $\simeq_1$.

Lemma 8.8. *$\alpha \simeq_1 \beta$ if and only if*

$$\exists \varphi : \alpha \longrightarrow \beta \; \exists \psi : \beta \longrightarrow \alpha \;\; \psi \circ_{n-1} \varphi \simeq_1 \mathrm{id}\, \alpha \;\wedge\; \varphi \circ_{n-1} \psi \simeq_1 \mathrm{id}\, \beta.$$

9 CONCLUSION

In this paper, we introduced an alternative definition of an ∞-category as a model of a certain typed equational theory. We showed the strong version of the theory, in which equations are identity, is equivalent to the established concept of strict ω-categories. A central motivation of the paper is constructive and mechanizable reasoning. For the free model we give a deterministic normalization procedure, yielding an effective decision procedure for equality of closed terms and supporting rewriting based automation. The same syntax also supports a weak reading in which equality is interpreted as equivalence witnessed by higher cells. We interpret this weak equality as homotopy relative to boundary, and we give a concrete homotopical semantics in which cells are boundary preserving homotopies and equations are inhabited by higher cells, so that the structural laws hold up to coherent equivalence. We develop a proof calculus for typing, composition, exchange, and identities, and establish soundness for this semantics.

Several directions remain. First, while dimensions are conventionally indexed by the natural numbers, our axioms do not fundamentally rely on linearity: the same presentation can be parameterized by a tree-ordered index structure. We aim to clarify what this added branching buys by exhibiting constructions and decision problems that are naturally expressed over tree orders, and that are not faithfully captured by purely linear indexing. Second, we would like to broaden the space of weak models and to isolate a minimal, abstract interface for *witnessed equivalence* sufficient for soundness of the proof calculus, beyond the concrete homotopy model developed here. Third, we plan to connect the weak reading more systematically to homotopy type theory (HoTT) [20]. HoTT treats types as spaces and equalities as paths, with semantics formulated in terms of $(\infty, 1)$-categories. We intend to identify which fragment of HoTT is validated by our weak equational theory, and to make precise how its derivations correspond to path algebra.

REFERENCES

[1] John C. Baez and James Dolan. 1998. Higher-Dimensional Algebra III.n-Categories and the Algebra of Opetopes. *Advances in Mathematics* 135, 2 (1998), 145–206. doi:10.1006/aima.1997.1695
[2] Michael Batanin. 1998. Monoidal globular categories as a natural environment for the theory of weak n-categories. *Advances in Mathematics* 136, 1 (1998), 39–103. doi:10.1006/aima.1998.1724
[3] Julia E. Bergner. 2006. A survey of (∞, 1)-categories. arXiv:math/0610239 [math.AT] https://arxiv.org/abs/math/0610239
[4] Julia E. Bergner. 2009. Complete Segal Spaces Arising from Simplicial Categories. *Trans. Amer. Math. Soc.* 361, 1 (2009), 525–546. http://www.jstor.org/stable/40302777
[5] Peter Booth. 2006. An explicit classification of three-stage Postnikov towers. *Homology, Homotopy and Applications* 8, 2 (01 2006), 133–155. doi:10.4310/HHA.2006.v8.n2.a8
[6] Ronald Brown and Philip J. Higgins. 1981. The equivalence of ∞-groupoids and crossed complexes. *Cahiers de topologie et géométrie différentielle* 22, 4 (1981), 371–386. https://www.numdam.org/item/CTGDC_1981__22_4_371_0/
[7] Denis-Charles Cisinski. 2019. *Higher categories and homotopical algebra.* Vol. 180. Cambridge University Press.
[8] Claudio Hermida, Michael Makkai, and John Power. 2002. On weak higher-dimensional categories I—3. *Journal of Pure and Applied Algebra* 166, 1 (2002), 83–104. doi:10.1016/S0022-4049(01)00014-7
[9] Andre Joyal and Myles Tierney. 2006. Quasi-categories vs Segal spaces. *Contemp. Math.* 431 (08 2006). doi:10.1090/conm/431/08278
[10] Donald Kahn. 1963. Induces Maps for Postnikov Systems. *Trans. Amer. Math. Soc.* 107, 3 (03 1963), 432–450. doi:10.2307/1993811
[11] Claude Kirchner and Hélène Kirchner. 2014. Equational logic and rewriting. In *Computational logic.* Handb. Hist. Log., Vol. 9. Elsevier/North-Holland, Amsterdam, 255–282. doi:10.1016/B978-0-444-51624-4.50006-X
[12] Markus Land. 2021. *Introduction to infinity-categories.* Springer Nature.
[13] Tom Leinster. 2004. *Higher operads, higher categories.* Number 298. Cambridge University Press.
[14] Jacob Lurie. 2009. *Higher topos theory.* Princeton University Press.
[15] V. Manca, A. Salibra, and G. Scollo. 1990. Equational type logic. *Theoretical Computer Science* 77, 1 (1990), 131–159. doi:10.1016/0304-3975(90)90118-2
[16] Mikhail M. Postnikov. 1951. Determination of the homology groups of a space by means of the homotopy invariants. *Doklady Akademii Nauk SSSR* 76 (1951), 359–362.
[17] Charles Rezk. 2001. A model for the homotopy theory of homotopy theory. *Trans. Amer. Math. Soc.* 353 (01 2001). doi:10.2307/221843
[18] Andrea Sattler-Klein. 1997. New undecidability results for finitely presented monoids. In *Rewriting Techniques and Applications (RTA'97) (Lecture Notes in Computer Science, 1232)*, Hubert Comon (Ed.). Springer, Berlin, Heidelberg, 68–82.
[19] Ross Street. 1987. The algebra of oriented simplexes. *Journal of Pure and Applied Algebra* 49, 3 (1987), 283–335. doi:10.1016/0022-4049(87)90137-X
[20] The Univalent Foundations Program. 2013. *Homotopy Type Theory: Univalent Foundations of Mathematics.* https://homotopytypetheory.org/book, Institute for Advanced Study.

A PROOFS FOR SECTION 4 (DECIDABILITY IN THE FREE MODEL)

Lemma 4.4 (Bottom-up composition on normal forms). *Let $u = \mathsf{id}^\ell \, \mathsf{end}_1^m a$ and $v = \mathsf{id}^i \, \mathsf{end}_2^j b$ be normal forms and let $\dim u = \dim v = n \geq k$. In the free model, the composition $u \circ_{n-k} v$ is typeable iff $a = b$ and one of the following holds:*

(a) *$k \leq \ell$ and $k \leq i$ and $u = v$ (equivalently, the conditions of Lemma 4.2 hold); in this case $u \circ_{n-k} v = u$.*
(b) *$k \leq \ell$ and $i < k$ and $\ell = k$ and $\mathsf{end}_1 = \mathsf{tgt}$ and $m = j + k - i$; in this case $u \circ_{n-k} v = u$.*
(c) *$\ell < k$ and $k \leq i$ and $i = k$ and $\mathsf{end}_2 = \mathsf{src}$ and $j = m + k - \ell$; in this case $u \circ_{n-k} v = v$.*

If $\ell < k$ and $i < k$ then the composition is not typeable in the free model.

Proof. A composition lowest in the tree is of the form

$$\mathsf{id}^\ell \, \mathsf{end}_1^m a \circ_{n-k} \mathsf{id}^i \, \mathsf{end}_2^j b \tag{15}$$

where $\ell, m, i, j \geq 0$, $\mathsf{end}_1, \mathsf{end}_2 \in \{\mathsf{src}, \mathsf{tgt}\}$, and a, b are primitive symbols. By well-formedness, we have

$$m \leq \dim a \qquad\qquad j \leq \dim b \qquad\qquad 1 \leq k \leq \dim a - m + \ell = \dim b - j + i.$$

The condition for the existence of the composition (15) is

$$\mathsf{src}^k \, \mathsf{id}^\ell \, \mathsf{end}_1^m a = \mathsf{tgt}^k \, \mathsf{id}^i \, \mathsf{end}_2^j b.$$

We can immediately declare failure if $a \neq b$, as these expressions cannot be equal in the free model. Thus the condition becomes

$$\mathsf{src}^k \, \mathsf{id}^\ell \, \mathsf{end}_1^m a = \mathsf{tgt}^k \, \mathsf{id}^i \, \mathsf{end}_2^j a \tag{16}$$

and we have

$$m, j \leq \dim a \qquad\qquad \ell - m = i - j.$$

Now we consider four cases depending on the ordering relation between ℓ and k and between i and k.

(i) $k \leq \ell$ and $k \leq i$
(ii) $k \leq \ell$ and $k > i$
(iii) $k > \ell$ and $k \leq i$
(iv) $k > \ell$ and $k > i$.

In case (i), using (R6)-(R9), the condition (16) becomes

$$\mathsf{id}^{\ell-k} \, \mathsf{end}_1^m a = \mathsf{id}^{i-k} \, \mathsf{end}_2^j a.$$

If $\ell \neq i$, say without loss of generality $\ell > i$, then applying src^{i-k} to both sides, we would have

$$\mathsf{id}^{\ell-i} \, \mathsf{end}_1^m a = \mathsf{end}_2^j a,$$

which does not hold in the free model, as the left-hand side is an identity and the right-hand side is not, so we can declare failure. Otherwise, if $\ell = i$, we would also have $j = m$, and the condition (16) becomes

$$\mathsf{id}^{\ell-k} \, \mathsf{end}_1^m a = \mathsf{id}^{\ell-k} \, \mathsf{end}_2^m a.$$

If $\mathsf{end}_1 \neq \mathsf{end}_2$ and $m > 0$, we can again declare failure, as the terms are not equal in the free model. Otherwise, the condition (16) becomes

$$\mathsf{id}^{\ell-k} \, \mathsf{end}^m a = \mathsf{id}^{\ell-k} \, \mathsf{end}^m a,$$

where end is either end_1 or end_2. These are equal, so we can form the composition (15) in the free model, which under the current assumptions becomes

$$\mathsf{id}^\ell \, \mathsf{end}^m a \circ_{n-k} \mathsf{id}^\ell \, \mathsf{end}^m a,$$

But this is equal to $\mathsf{id}^\ell \, \mathsf{end}^m a$:

$$\begin{aligned}
&\mathsf{id}^\ell \, \mathsf{end}^m a \circ_{n-k} \mathsf{id}^\ell \, \mathsf{end}^m a \\
&= \mathsf{id}^k \, \mathsf{id}^{\ell-k} \, \mathsf{end}^m a \circ_{n-k} \mathsf{id}^\ell \, \mathsf{end}^m a && \text{since } \ell \geq k \\
&= \mathsf{id}^k \, \mathsf{tgt}^k \, \mathsf{id}^k \, \mathsf{id}^{\ell-k} \, \mathsf{end}^m a \circ_{n-k} \mathsf{id}^\ell \, \mathsf{end}^m a && \text{by (R7)} \\
&= \mathsf{id}^k \, \mathsf{tgt}^k \, \mathsf{id}^\ell \, \mathsf{end}^m a \circ_{n-k} \mathsf{id}^\ell \, \mathsf{end}^m a \\
&= \mathsf{id}^\ell \, \mathsf{end}^m a && \text{by (R11),}
\end{aligned}$$

thus we can replace (15) with $\mathsf{id}^\ell \, \mathsf{end}^m a$ and we have eliminated a composition.

The cases (ii) and (iii) are symmetric, so we only do (ii). Again using (R6)-(R9), the condition (16) becomes

$$\mathsf{id}^{\ell-k} \, \mathsf{end}_1^m a = \mathsf{tgt}^{k-i+j} \, a.$$

If $k < \ell$, then this condition does not hold in the free model, as the left-hand side is an identity and the right-hand side is not, so we can declare failure. Otherwise $\ell = k$ and $m = k - i + j$, and the condition (16) becomes

$$\mathsf{end}_1^m a = \mathsf{tgt}^m\, a.$$

If $\mathsf{end}_1 \neq \mathsf{tgt}$ and $m > 0$, we can again declare failure, as the terms are not equal in the free model. Otherwise, our condition becomes

$$\mathsf{end}^m a = \mathsf{end}^m a,$$

where end = tgt. These are equal, so we can form the composition (15) in the free model, which under the current assumptions becomes

$$\mathsf{id}^\ell\, \mathsf{end}^m a \circ_{n-k} \mathsf{id}^\ell\, \mathsf{end}^m a.$$

The remainder of the argument is as in case (i).

Finally, in case (iv), using (R6)-(R9), the condition (16) becomes

$$\mathsf{src}^{k-\ell+m}\, a = \mathsf{tgt}^{k-i+j}\, a,$$

which does not hold in the free model as $k - \ell + m > 0$ and $k - i + j > 0$, so we can declare failure. □

B PROOFS FOR SECTION 6 (ADEQUACY OF STRICT ω-CATEGORIES)

Theorem 6.1. *An* EqCat_∞ *model is exactly a strict* ω*-category.*

Proof. ($\Rightarrow$) Assume we have a model C of EqCat_∞. Let $x \in C_k$ for $k \geq 0$. Then:

$$\mathsf{src}\,\mathsf{src}\, x = \mathsf{src}\,\mathsf{tgt}\, x$$
$$\mathsf{tgt}\,\mathsf{src}\, x = \mathsf{tgt}\,\mathsf{tgt}\, x$$

by (IC1) and (IC2). Thus, $(\cdots C_2 \rightrightarrows C_1 \rightrightarrows C_0)$ forms a globular set.

We now show for all $k > \ell > m$, $(C_k \rightrightarrows C_\ell \rightrightarrows C_m)$ forms a strict 2-category, where objects will be m-cells, morphisms ℓ-cells, and 2-morphisms k-cells. Furthermore, note $\mathsf{src}^{\ell-m}, \mathsf{tgt}^{\ell-m} \colon C_\ell \to C_m$, and $\mathsf{src}^{k-\ell}, \mathsf{tgt}^{k-\ell} \colon C_k \to C_\ell$.

Each object $a \in C_m$ has an identity; namely, $\mathsf{id}^{\ell-m} \colon a \to a$ (since $\mathsf{src}^{\ell-m}\, \mathsf{id}^{\ell-m}\, a = a$ and $\mathsf{tgt}^{\ell-m}\, \mathsf{id}^{\ell-m}\, a = a$.) Furthermore, if $f \colon a \to b$ and $g \colon b \to c$ in C_ℓ, then $g \circ_m f \colon a \to c$. This is because by repeated application of (IC5),

$$\mathsf{src}^{\ell-m}(g \circ_m f) = \mathsf{src}((\mathsf{src}^{\ell-(m+1)}\, g) \circ_m (\mathsf{src}^{\ell-(m+1)}\, f)) = \mathsf{src}\,\mathsf{src}^{\ell-(m+1)}\, f = \mathsf{src}^{\ell-m}\, f = a,$$

and similarly $\mathsf{tgt}^{\ell-m}(g \circ_m f) = c$. Associativity of composition follows directly from (IC9).

For the 2-category structure, for $f \colon a \to b$, note that $\mathsf{id}^{k-\ell} f \colon f \to f$ since $\mathsf{src}^{k-\ell}\, \mathsf{id}^{k-\ell} f = f$ and $\mathsf{tgt}^{k-\ell}\, \mathsf{id}^{k-\ell} f = f$. For $\alpha \colon f \to g$ and $\beta \colon g \to h$, $\beta \circ_\ell \alpha \colon f \to h$ and this is associative, following from (IC5), (IC6), and (IC9), respectively.

Finally, to show the exchange property would amount to showing for $\alpha, \gamma, \beta, \sigma \in C_k$,

$$(\sigma \circ_m \beta) \circ_\ell (\gamma \circ_m \alpha) = (\sigma \circ_\ell \gamma) \circ_m (f \circ_\ell \alpha)$$

which follows directly from (IC10).

($\Leftarrow$) Here. For the reverse direction, assume we have a strict ω−category. We show the axioms of EqCat_∞ are satisfied.

Since $(\cdots \rightrightarrows C_2 \rightrightarrows C_1 \rightrightarrows C_0)$ forms a globular set, (IC1) and (IC2) hold. (IC3) and (IC4) hold by viewing $C_n \rightrightarrows C_{n-1}$ as a category.

Furthermore, (IC5) and (IC6) hold by viewing ...finish this one..

(IC7) and (IC8):

$$\mathsf{src}(y \circ_n x) = \mathsf{src}\, y \circ_n \mathsf{src}\, x$$
$$\mathsf{tgt}(y \circ_n x) = \mathsf{tgt}\, y \circ_n \mathsf{tgt}\, x$$

where $\dim x = \dim y = d > n + 1$, are obtained from viewing $C_d \rightrightarrows C_{d-1} \rightrightarrows C_n$ as a strict 2-category: let $x \colon f \to g$ and $y \colon f' \to g'$ be in C_d. Then $\mathsf{src}(y \circ_n x) = f' \circ_n f$ and $\mathsf{src}\, y \circ_n \mathsf{src}\, x = f' \circ_n f$ which provides the desired equality. (IC8) follows the exact same argument.

For $\dim x = \dim y = \dim z = d > n$, (IC9) follows from morphism associativity by viewing $C_d \rightrightarrows C_n$ as a category.

For $d > m > n$, (IC10):

$$(x \circ_n y) \circ_m (z \circ_n w) = (x \circ_m z) \circ_n (y \circ_m w)$$

is a consequence of the exchange rule for the strict 2-category $C_d \rightrightarrows C_m \rightrightarrows C_n$. In addition,

$$\mathsf{src}^{d-m}\, \mathsf{id}^{m-d}\, x = x \qquad\qquad x \circ_n \mathsf{id}^{n-d}\, \mathsf{src}^{d-n}\, x = x$$
$$\mathsf{tgt}^{d-m}\, \mathsf{id}^{m-d}\, x = x \qquad\qquad \mathsf{id}^{n-d}\, \mathsf{tgt}^{d-n}\, x \circ_n x = x$$

are immediate upon viewing $\mathsf{id} = \mathsf{id}^{m-d}$ in the context of $C_d \rightrightarrows C_m$ as a category, and so (IC11) and (IC12) are satisfied. Lastly, if $n = \dim x = \dim y > k$, then (IC13) follows from viewing $C_{n+1} \rightrightarrows C_n \rightrightarrows C_k$ as a strict 2-category.

□

C PROOFS FOR SECTION 7 (HOMOTOPY MODELS)

Lemma 7.1. *Let $\alpha \in H_n$. The following are equivalent:*

(i) *α is a cell;*
(ii) *for all $2 \leq k \leq n$, for all $x_1, \ldots, x_{k-1} \in I^{k-1}$ and $b \in \{0,1\}$,*

$$\alpha^{(1,2,\ldots,k)}(x_1, \ldots, x_{k-1}, b) = \alpha^{(1,2,\ldots,k)}(b, \ldots, b, b); \tag{11}$$

(iii) *for all $1 \leq k \leq n$, $\alpha^{(k)}(0) = \mathrm{id}^{k-1} \operatorname{src}^k \alpha$ and $\alpha^{(k)}(1) = \mathrm{id}^{k-1} \operatorname{tgt}^k \alpha$;*
(iv) *for all $1 \leq m \leq n-1$ and $b \in \{0,1\}$, the value of $\alpha^{(m,m+1)}(x, b)$ is independent of x.*

Proof. First we observe that if α is a cell, then so are $\operatorname{src} \alpha$, $\operatorname{tgt} \alpha$, and $\mathrm{id}\, \alpha$. The first two are by definition of cells. For $\mathrm{id}\,\alpha$, we have $(\mathrm{id}\,\alpha)^{(1)}(t) = \alpha$, so the first condition is satisfied, and $(\mathrm{id}\,\alpha)^{(2)}(b)^{(1)}(t) = (\mathrm{id}\,\alpha)^{(1)}(t)^{(1)}(b) = \alpha^{(1)}(b)$, which is independent of t, so the second condition is satisfied.

(i) $\Rightarrow$ (ii): The argument is by induction on dimension.

$$\begin{aligned}
&\alpha^{(1,\ldots,k)}(x_1, \ldots, x_{k-1}, b) \\
&= \alpha^{(1)}(x_1)^{(1,\ldots,k-1)}(x_2, \ldots, x_{k-1}, b) \\
&= \alpha^{(1)}(x_1)^{(1,\ldots,k-1)}(b, \ldots, b, b) && \text{induction hypothesis} \\
&= \alpha^{(2)}(b)^{(1)}(x_1)^{(1,\ldots,k-2)}(b, \ldots, b) \\
&= \alpha^{(2)}(b)^{(1)}(b)^{(1,\ldots,k-2)}(b, \ldots, b) && \text{since } \alpha^{(2)}(b) \text{ is an identity} \\
&= \alpha^{(1,\ldots,k)}(b, \ldots, b).
\end{aligned}$$

(ii) $\Rightarrow$ (iii): By induction on k.

The base case $k = 1$ is by definition of src. For $k \geq 2$,

$$\begin{aligned}
\alpha^{(k)}(0)^{(1)}(t) &= \alpha^{(1,k)}(t, 0) = \alpha^{(1,k)}(0,0) \\
&= \alpha^{(1)}(0)^{(k-1)}(0) = (\operatorname{src} \alpha)^{(k-1)}(0)
\end{aligned}$$

with the second inference from (11). By the induction hypothesis, this is $\mathrm{id}^{k-2} \operatorname{src}^{k-1} \operatorname{src} \alpha = \mathrm{id}^{k-2} \operatorname{src}^k \alpha$. As this is independent of t, $\alpha^{(k)}(0) = \mathrm{id}^{k-1} \operatorname{src}^k \alpha$. The argument for $\alpha^{(k)}(1) = \mathrm{id}^{k-1} \operatorname{tgt}^k \alpha$ is symmetric.

(iii) $\Rightarrow$ (iv): For all $1 \leq m \leq n-1$, for all $x_1, \ldots, x_m, x$,

$$\begin{aligned}
&\alpha^{(m,m+1)}(x, 0)^{(1,2,\ldots,m-1)}(x_1, \ldots, x_{m-1}) \\
&= \alpha^{(m+1)}(0)^{(1,2,\ldots,m)}(x_1, \ldots, x_{m-1}, x) \\
&= (\mathrm{id}^m \operatorname{src}^{m+1} \alpha)^{(1,2,\ldots,m)}(x_1, \ldots, x_{m-1}, x) = \operatorname{src}^{m+1} \alpha,
\end{aligned}$$

which is independent of x, therefore so is $\alpha^{(m,m+1)}(x, 0)$. Symmetrically, the same holds for $\alpha^{(m,m+1)}(x, 1)$.

(iv) $\Rightarrow$ (i): By induction on dimension. For α satisfying (iv), $1 \leq m \leq n-2$, $t, x \in I$, and $b \in \{0,1\}$,

$$\begin{aligned}
\alpha^{(1)}(t)^{(m,m+1)}(x, b) &= \alpha^{(m+1,m+2)}(x, b)^{(1)}(t) \\
&= \alpha^{(m+1,m+2)}(0, b)^{(1)}(t) = \alpha^{(1)}(t)^{(m,m+1)}(0, b),
\end{aligned}$$

so $\alpha^{(1)}(t)$ satisfies (iii). By the induction hypothesis, $\alpha^{(1)}(t)$ is a cell. Moreover, $\alpha^{(1,2)}(t, b)$ is independent of t by assumption, thus $\alpha^{(2)}(b)$ is an identity. Therefore α is a cell. □

Lemma 7.2. *All sections of cells are cells.*

Proof. By induction on dimension. Any section of dimension zero or one is a cell by definition. For $n \geq 2$ and $1 \leq i_1 < \cdots < i_k \leq n$,

$$\alpha^{(i_1,\ldots,i_k)}(x_{i_1}, \ldots, x_{i_k})^{(1)}(x) = \alpha^{(j_1,\ldots,j_{k+1})}(x_{j_1}, \ldots, x_{j_{k+1}})$$

after reindexing, which is a cell by the induction hypothesis, thus $\alpha^{(i_1,\ldots,i_k)}(x_{i_1}, \ldots, x_{i_k})$ satisfies the first condition of cellhood. For the second condition, we must show that $\alpha^{(i_1,\ldots,i_k)}(x_{i_1}, \ldots, x_{i_k})^{(2)}(0)$ is an identity, provided $n - k \geq 2$ (otherwise, there is nothing to prove). By (11),

$$\begin{aligned}
\alpha^{(i_1,\ldots,i_k)}(x_{i_1}, \ldots, x_{i_k})^{(2)}(b)^{(1)}(x) &= \alpha^{(i_1,\ldots,i_k)}(x_{i_1}, \ldots, x_{i_k})^{(1,2)}(x, b) \\
&= \alpha^{(i_1,\ldots,i_k)}(x_{i_1}, \ldots, x_{i_k})^{(1,2)}(b, b).
\end{aligned}$$

As this is independent of x, $\alpha^{(i_1,\ldots,i_k)}(x_{i_1},\ldots,x_{i_k})^{(2)}(b)$ is an identity. □

Lemma 7.4. *For $\alpha \in C_n$, $f : I^2 \to I$, $x, y \in I$, and $1 \le k \le n$,*

$$\begin{aligned}&\mathsf{rpm}(\alpha,f,k)^{(m+1,m+2)}(x,y)\\ &= \begin{cases}\mathsf{rpm}(\alpha^{(m,m+1)}(x,y),f,k-2) & \text{if } 1 \le m \le k-2\\ \mathsf{rpm}(\alpha^{(m,m+1)}(x,y),f,k) & \text{if } k+1 \le m \le n-1.\end{cases}\end{aligned}$$

For the remaining two values of m not covered above,

$$\mathsf{rpm}(\alpha,f,k)^{(k,k+1)}(x,s)^{(1)}(t) = \alpha^{(k-1,k)}(x,f(t,s))$$
$$\mathsf{rpm}(\alpha,f,k)^{(k+1,k+2)}(s,y)^{(1)}(t) = \alpha^{(k,k+1)}(f(t,s),y).$$

Proof. For the case $1 \le m \le k-2$,

$$\begin{aligned}&\mathsf{rpm}(\alpha,f,k)^{(m+1,m+2)}(x,y)^{(1,k-1)}(t,s)\\ &= \mathsf{rpm}(\alpha,f,k)^{(1,k+1)}(t,s)^{(m,m+1)}(x,y)\\ &= \alpha^{(k)}(f(t,s))^{(m,m+1)}(x,y)\\ &= \alpha^{(m,m+1)}(x,y)^{(k-2)}(f(t,s))\\ &= \mathsf{rpm}(\alpha^{(m,m+1)}(x,y),f,k-2)^{(1,k-1)}(t,s),\end{aligned}$$

thus $\mathsf{rpm}(\alpha,f,k)^{(m+1,m+2)}(x,y) = \mathsf{rpm}(\alpha^{(m,m+1)}(x,y),f,k-2)$.

For $k+1 \le m \le n-1$,

$$\begin{aligned}&\mathsf{rpm}(\alpha,f,k)^{(m+1,m+2)}(x,y)^{(1,k+1)}(t,s)\\ &= \mathsf{rpm}(\alpha,f,k)^{(1,k+1)}(t,s)^{(m-1,m)}(x,y)\\ &= \alpha^{(k)}(f(t,s))^{(m-1,m)}(x,y)\\ &= \alpha^{(m,m+1)}(x,y)^{(k)}(f(t,s))\\ &= \mathsf{rpm}(\alpha^{(m,m+1)}(x,y),f,k)^{(1,k+1)}(t,s),\end{aligned}$$

thus $\mathsf{rpm}(\alpha,f,k)^{(m+1,m+2)}(x,y) = \mathsf{rpm}(\alpha^{(m,m+1)}(x,y),f,k)$.

For the remaining two cases,

$$\begin{aligned}&\mathsf{rpm}(\alpha,f,k)^{(k,k+1)}(x,s)^{(1)}(t)\\ &= \mathsf{rpm}(\alpha,f,k)^{(1,k+1)}(t,s)^{(k-1)}(x)\\ &= \alpha^{(k)}(f(t,s))^{(k-1)}(x) = \alpha^{(k-1,k)}(x,f(t,s))\end{aligned}$$

$$\begin{aligned}&\mathsf{rpm}(\alpha,f,k)^{(k+1,k+2)}(s,y)^{(1)}(t)\\ &= \mathsf{rpm}(\alpha,f,k)^{(1,k+1,k+2)}(t,s,y)\\ &= \mathsf{rpm}(\alpha,f,k)^{(1,k+1)}(t,s)^{(k)}(y)\\ &= \alpha^{(k)}(f(t,s))^{(k)}(y) = \alpha^{(k,k+1)}(f(t,s),y).\end{aligned}$$

□

Lemma 7.5. *Let $f : I^2 \to I$ be a continuous function such that for $b \in \{0,1\}$, $f^{(2)}(b)$ is a constant function* $\mathsf{id}\, c$ *for some $c \in \{0,1\}$. Then for all $\alpha \in C_n$ and $1 \le k \le n$, $\mathsf{rpm}(\alpha,f,k) \in C_{n+1}$.*

Proof. We have already argued above that $\mathsf{rpm}(\alpha,f,k) \in H_{n+1}$. By Lemma 7.1(iv), it suffices to show that all $\mathsf{rpm}(\alpha,f,k)^{(m+1,m+2)}(x,b)$ are independent of x for $b \in \{0,1\}$. Using Lemma 7.4,

$$\begin{aligned}&\mathsf{rpm}(\alpha,f,k)^{(m+1,m+2)}(x,b)\\ &= \begin{cases}\mathsf{rpm}(\alpha^{(m,m+1)}(x,b),f,k-2) & \text{if } 1 \le m \le k-2\\ \mathsf{rpm}(\alpha^{(m,m+1)}(x,b),f,k) & \text{if } k+1 \le m \le n-1\end{cases}\\ &\mathsf{rpm}(\alpha,f,k)^{(k,k+1)}(x,b)^{(1)}(t) = \alpha^{(k-1,k)}(x,f(t,b))\\ &\mathsf{rpm}(\alpha,f,k)^{(k+1,k+2)}(x,b)^{(1)}(t) = \alpha^{(k,k+1)}(f(t,x),b).\end{aligned}$$

By assumption, $f(t,b) = c \in \{0,1\}$. Since α is a cell, the expressions on the right-hand sides of all these equations are independent of x by Lemma 7.1(iv). ◻

Lemma 7.7. *The relation $\simeq_k$ is an equivalence relation on C_n^k and the relations $\simeq_k$ form a stratified congruence with respect to the operations* $\mathrm{src}, \mathrm{tgt} : C_n \to C_{n-1}$ *for $n \geq 1$,* $\mathrm{id} : C_n \to C_{n+1}$*, and $\circ_m \colon C_n \times C_n \to C_n$ for $0 \leq m \leq n-1$ in the following sense: for $\alpha, \beta \in C_n^k$,*

(i) *If $\alpha \simeq_k \beta$ and $k \geq 2$, then $\mathrm{src}\,\alpha \simeq_{k-1} \mathrm{src}\,\beta$ and $\mathrm{tgt}\,\alpha \simeq_{k-1} \mathrm{tgt}\,\beta$.*
(ii) *If $\alpha \simeq_1 \beta$, then $\mathrm{src}\,\alpha = \mathrm{src}\,\beta$ and $\mathrm{tgt}\,\alpha = \mathrm{tgt}\,\beta$.*
(iii) *If $\alpha \simeq_k \beta$, then $\mathrm{id}\,\alpha \simeq_{k+1} \mathrm{id}\,\beta$.*
(iv) *For $m \geq k$, if $\alpha \simeq_k \alpha'$ and $\beta \simeq_k \beta'$, and if $\beta \circ_{n-m} \alpha$ and $\beta' \circ_{n-m} \alpha'$ exist, then $\beta \circ_{n-m} \alpha \simeq_k \beta' \circ_{n-m} \alpha'$.*

Proof. The relations $\simeq_k$ are reflexive, as $\mathrm{id}\,\alpha$ witnesses $\alpha \simeq_k \alpha$ whenever $\alpha \in C_n^k$. If $\alpha = \mathrm{id}^{k-1}\,\mathrm{src}^{k-1}\,\alpha = \mathrm{id}^{k-1}\,\mathrm{tgt}^{k-1}\,\alpha$, then by Lemma 7.1,

$$(\mathrm{id}\,\alpha)^{(k)}(0) = \mathrm{id}^{k-1}\,\mathrm{src}^{k}\,\mathrm{id}\,\alpha = \mathrm{id}^{k-1}\,\mathrm{src}^{k-1}\,\alpha = \alpha$$
$$(\mathrm{id}\,\alpha)^{(k)}(1) = \mathrm{id}^{k-1}\,\mathrm{tgt}^{k}\,\mathrm{id}\,\alpha = \mathrm{id}^{k-1}\,\mathrm{tgt}^{k-1}\,\alpha = \alpha.$$

They are symmetric, as the witnessses must be invertible.

They are transitive, as $\psi \circ_{n-k} \varphi$ witnesses $\gamma \simeq_k \alpha$ when φ witnesses $\alpha \simeq_k \beta$ and ψ witnesses $\beta \simeq_k \gamma$. Since $\psi^{(k)}(0) = \beta = \varphi^{(k)}(1)$, the composition $\psi \circ_{n-k} \varphi$ exists, and by (9),

$$\begin{aligned}(\psi \circ_{n-k} \varphi)^{(k)}(0) &= \mathrm{id}^{k-1}\,\mathrm{src}^{k}(\psi \circ_{n-k} \varphi)\\ &= \mathrm{id}^{k-1}\,\mathrm{src}^{k}\,\varphi = \varphi^{(k)}(0) = \alpha\\ (\psi \circ_{n-k} \varphi)^{(k)}(1) &= \mathrm{id}^{k-1}\,\mathrm{tgt}^{k}(\psi \circ_{n-k} \varphi)\\ &= \mathrm{id}^{k-1}\,\mathrm{tgt}^{k}\,\psi = \psi^{(k)}(1) = \gamma.\end{aligned}$$

For (i), suppose φ witnesses $\alpha \simeq_k \beta$, so $\varphi^{(k)}(0) = \alpha$ and $\varphi^{(k)}(1) = \beta$. Then

$$\begin{aligned}(\mathrm{src}\,\varphi)^{(k-1)}(0) &= \varphi^{(1)}(0)^{(k-1)}(0) = \varphi^{(k)}(0)^{(1)}(0)\\ &= \mathrm{src}\,\varphi^{(k)}(0) = \mathrm{src}\,\alpha\\ (\mathrm{src}\,\varphi)^{(k-1)}(1) &= \varphi^{(1)}(0)^{(k-1)}(1) = \varphi^{(k)}(1)^{(1)}(0)\\ &= \mathrm{src}\,\varphi^{(k)}(1) = \mathrm{src}\,\beta,\end{aligned}$$

so $\mathrm{src}\,\varphi$ witnesses $\mathrm{src}\,\alpha \simeq_{k-1} \mathrm{src}\,\beta$. Similarly, $\mathrm{tgt}\,\varphi$ witnesses $\mathrm{tgt}\,\alpha \simeq_{k-1} \mathrm{tgt}\,\beta$.

For (ii), if φ witnesses $\alpha \simeq_1 \beta$, then $\alpha = \mathrm{src}\,\varphi$ and $\beta = \mathrm{tgt}\,\varphi$. Then

$$\mathrm{src}\,\alpha = \mathrm{src}^2\,\varphi = \mathrm{src}\,\mathrm{tgt}\,\varphi = \mathrm{src}\,\beta \qquad\qquad \mathrm{tgt}\,\alpha = \mathrm{tgt}\,\mathrm{src}\,\varphi = \mathrm{tgt}^2\,\varphi = \mathrm{tgt}\,\beta.$$

For (iii), suppose φ witnesses $\alpha \simeq_k \beta$. By Lemma 7.1, $\alpha = \varphi^{(k)}(0) = \mathrm{id}^{k-1}\,\mathrm{src}^k\,\varphi$ and $\beta = \varphi^{(k)}(1) = \mathrm{id}^{k-1}\,\mathrm{tgt}^k\,\varphi$.

$$\begin{aligned}(\mathrm{id}\,\varphi)^{(k+1)}(0) &= \mathrm{id}^{k}\,\mathrm{src}^{k+1}\,\mathrm{id}\,\varphi = \mathrm{id}^{k}\,\mathrm{src}^{k}\,\varphi = \mathrm{id}\,\mathrm{id}^{k-1}\,\mathrm{src}^{k}\,\varphi\\ &= \mathrm{id}\,\varphi^{(k)}(0) = \mathrm{id}\,\alpha\\ (\mathrm{id}\,\varphi)^{(k+1)}(1) &= \mathrm{id}^{k}\,\mathrm{tgt}^{k+1}\,\mathrm{id}\,\varphi = \mathrm{id}^{k}\,\mathrm{tgt}^{k}\,\varphi = \mathrm{id}\,\mathrm{id}^{k-1}\,\mathrm{tgt}^{k}\,\varphi\\ &= \mathrm{id}\,\varphi^{(k)}(1) = \mathrm{id}\,\beta,\end{aligned}$$

so $\mathrm{id}\,\varphi$ witnesses $\mathrm{id}\,\alpha \simeq_{k+1} \mathrm{id}\,\beta$.

For (iv), suppose $m \geq k$. Let φ witness $\alpha \simeq_k \alpha'$, so $\alpha = \varphi^{(k)}(0) = \mathrm{id}^{k-1} = \mathrm{id}^{k-1}\,\mathrm{src}^k\,\varphi$ and $\alpha' = \varphi^{(k)}(1) = \mathrm{id}^{k-1}\,\mathrm{tgt}^k\,\varphi$. Let ψ witness $\beta \simeq_k \beta'$, so $\beta = \psi^{(k)}(0) = \mathrm{id}^{k-1}\,\mathrm{src}^k\,\psi$ and $\beta' = \psi^{(k)}(1) = \mathrm{id}^{k-1}\,\mathrm{id}^{k-1}\,\mathrm{tgt}^k\,\psi$. We have

$$\psi^{(m+1)}(0)^{(k)}(0) = \psi^{(k)}(0)^{(m)}(0) = \beta^{(m)}(0)$$
$$\varphi^{(m+1)}(1)^{(k)}(0) = \varphi^{(k)}(0)^{(m)}(1) = \alpha^{(m)}(1).$$

Since $\beta \circ_{n-m} \alpha$ exists, it must be that $\beta^{(m)}(0) = \alpha^{(m)}(1)$, therefore

$$\psi^{(m+1)}(0)^{(k)}(0) = \varphi^{(m+1)}(1)^{(k)}(0) \qquad\qquad \psi^{(m+1)}(0) = \varphi^{(m+1)}(1).$$

This implies that $\psi \circ_{n-m} \varphi$ exists, and it is the desired witness of $\beta \circ_{n-m} \alpha \simeq_k \beta' \circ_{n-m} \alpha'$, since by (9) and (10),

$$\begin{aligned}
&(\psi \circ_{n-m} \varphi)^{(k)}(0)\\
&= \mathrm{id}^{k-1} \mathrm{src}^k (\psi \circ_{n-m} \varphi) = \mathrm{id}^{k-1}(\mathrm{src}^k \psi \circ_{n-m} \mathrm{src}^k \varphi)\\
&= (\mathrm{id}^{k-1} \mathrm{src}^k \psi \circ_{n-m} \mathrm{id}^{k-1} \mathrm{src}^k \varphi) = \beta \circ_{n-m} \alpha\\
&(\psi \circ_{n-m} \varphi)^{(k)}(1)\\
&= \mathrm{id}^{k-1} \mathrm{tgt}^k (\psi \circ_{n-m} \varphi) = \mathrm{id}^{k-1}(\mathrm{tgt}^k \psi \circ_{n-m} \mathrm{tgt}^k \varphi)\\
&= (\mathrm{id}^{k-1} \mathrm{tgt}^k \psi \circ_{n-m} \mathrm{id}^{k-1} \mathrm{tgt}^k \varphi) = \beta' \circ_{n-m} \alpha'.
\end{aligned}$$ □

Lemma 7.8. *Let $\alpha, \beta \in C_n$ with $\alpha \simeq_k \beta$.*

(i) *If $\alpha, \beta \in C_n^m$ for some $m > k$, then $\alpha = \beta$.*
(ii) *If $\alpha \simeq_m \beta$ for some $m > k$, then $\alpha = \beta$.*

Proof. (i) Since $\alpha, \beta \in C_n^m$,

$$\alpha = \mathrm{id}^{m-1} \mathrm{src}^{m-1} \alpha = \mathrm{id}^{m-1} \mathrm{tgt}^{m-1} \alpha$$
$$\beta = \mathrm{id}^{m-1} \mathrm{src}^{m-1} \beta = \mathrm{id}^{m-1} \mathrm{tgt}^{m-1} \beta.$$

Moreover, if $\alpha \simeq_k \beta$ is witnessed by φ, then $\alpha = \mathrm{id}^{k-1} \mathrm{src}^k \varphi$ and $\beta = \mathrm{id}^{k-1} \mathrm{tgt}^k \varphi$, so

$$\begin{aligned}
\alpha &= \mathrm{id}^{m-1} \mathrm{src}^{m-1} \alpha = \mathrm{id}^{m-1} \mathrm{src}^{m-1} \mathrm{id}^{k-1} \mathrm{src}^k \varphi\\
&= \mathrm{id}^{m-1} \mathrm{src}^{m-k} \mathrm{src}^k \varphi = \mathrm{id}^{m-1} \mathrm{src}^m \varphi\\
\beta &= \mathrm{id}^{m-1} \mathrm{src}^{m-1} \beta = \mathrm{id}^{m-1} \mathrm{src}^{m-1} \mathrm{id}^{k-1} \mathrm{tgt}^k \varphi\\
&= \mathrm{id}^{m-1} \mathrm{src}^{m-k} \mathrm{tgt}^k \varphi = \mathrm{id}^{m-1} \mathrm{src}^m \varphi,
\end{aligned}$$

therefore $\alpha = \beta$.

(ii) This is immediate from (i), as $\alpha \simeq_m \beta$ implies $\alpha, \beta \in C_n^m$. □

Lemma 7.9. *The map $\mathrm{id} : C_n \to C_{n+1}$, restricted to C_n^k, is a bijection $\mathrm{id} : C_n^k \to C_{n+1}^{k+1}$ with two-sided inverse $\mathrm{src} = \mathrm{tgt} : C_{n+1}^{k+1} \to C_n^k$. For $\alpha, \beta \in C_{n+1}^{k+1}$, $\alpha \simeq_{k+1} \beta$ if and only if $\mathrm{src}\,\alpha \simeq_k \mathrm{src}\,\beta$, and also if and only if $\mathrm{tgt}\,\alpha \simeq_k \mathrm{tgt}\,\beta$.*

Proof. If $\alpha \in C_n^k$, then $\alpha = \mathrm{id}^{k-1} \mathrm{src}^{k-1} \alpha$, so

$$\mathrm{id}\,\alpha = \mathrm{id}\,\mathrm{id}^{k-1} \mathrm{src}^{k-1} \alpha = \mathrm{id}^k \mathrm{src}^k \mathrm{id}\,\alpha,$$

so $\mathrm{id}\,\alpha \in C_{n+1}^{k+1}$. Conversely, if $\alpha \in C_{n+1}^{k+1}$, then $\alpha = \mathrm{id}^k \mathrm{src}^k \alpha$, so $\mathrm{src}\,\alpha = \mathrm{src}\,\mathrm{id}^k \mathrm{src}^k \alpha = \mathrm{id}^{k-1} \mathrm{src}^{k-1} \mathrm{src}\,\alpha$, so $\mathrm{src}\,\alpha \in C_n^k$. Since $\mathrm{src}\,\mathrm{id}\,\alpha = \alpha$, src is a left inverse of id, but also a right inverse on elements of C_{n+1}^{k+1}: for $\alpha \in C_{n+1}^{k+1}$, $\mathrm{id}\,\mathrm{src}\,\alpha = \mathrm{id}\,\mathrm{src}\,\mathrm{id}^k \mathrm{src}^k \alpha = \mathrm{id}^k \mathrm{src}^k \alpha = \alpha$.

If $\alpha \simeq_{k+1} \beta$, then $\mathrm{src}\,\alpha \simeq_k \mathrm{src}\,\beta$ as observed in Lemma 7.7(i). Conversely, if $\alpha \simeq_k \beta$, then $\mathrm{id}\,\alpha \simeq_{k+1} \mathrm{id}\,\beta$ as observed in Lemma 7.7(iii).

The same arguments are valid with src replaced by tgt. □

D PROOFS FOR SECTION 8 (SOUNDNESS OF THE AXIOMS)

Lemma 8.1 (Associativity). *Let $1 \le k \le n$ and let α, β, γ be n-cells with $\alpha^{(k)}(1) = \beta^{(k)}(0)$ and $\beta^{(k)}(1) = \gamma^{(k)}(0)$. Then*

$$(\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha \simeq_1 \gamma \circ_{n-k} (\beta \circ_{n-k} \alpha).$$

Proof. This is done with reparameterization. We claim that

$$\begin{aligned}
&\mathrm{rpm}(\gamma \circ_{n-k} (\beta \circ_{n-k} \alpha), f^{\leftarrow}_{1/2,2/3}, k)\\
&\quad \circ_{n-1} \ \mathrm{rpm}((\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha, f_{1/2,1/3}, k)
\end{aligned} \tag{17}$$

is the desired witness. The composition (17) exists provided

$$\begin{aligned}
&\mathrm{tgt}\,\mathrm{rpm}((\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha, f_{1/2,1/3}, k)\\
&= \mathrm{src}\,\mathrm{rpm}(\gamma \circ_{n-k} (\beta \circ_{n-k} \alpha), f^{\leftarrow}_{1/2,2/3}, k)\\
&= \mathrm{tgt}\,\mathrm{rpm}(\gamma \circ_{n-k} (\beta \circ_{n-k} \alpha), f_{1/2,2/3}, k).
\end{aligned} \tag{18}$$

The targets occur at $t = 1$, giving

$$\begin{aligned}
&\mathsf{rpm}((\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha, f_{1/2,1/3}, k)^{(1,k+1)}(1, s) \\
&= ((\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha)^{(k)}(f_{1/2,1/3}(s)) \\
&= \begin{cases} \alpha^{(k)}(2f_{1/2,1/3}(s)), & \text{if } 0 \le f_{1/2,1/3}(s) \le 1/2 \\ \beta^{(k)}(4f_{1/2,1/3}(s) - 2), & \text{if } 1/2 \le f_{1/2,1/3}(s) \le 3/4 \\ \gamma^{(k)}(4f_{1/2,1/3}(s) - 3), & \text{if } 3/4 \le f_{1/2,1/3}(s) \le 1 \end{cases}
\end{aligned}$$

$$\begin{aligned}
&\mathsf{rpm}(\gamma \circ_{n-k} (\beta \circ_{n-k} \alpha), f_{1/2,2/3}, k)^{(1,k+1)}(1, s) \\
&= (\gamma \circ_{n-k} (\beta \circ_{n-k} \alpha))^{(k)}(f_{1/2,2/3}(s)) \\
&= \begin{cases} \alpha^{(k)}(4f_{1/2,2/3}(s)), & \text{if } 0 \le f_{1/2,2/3}(s) \le 1/4 \\ \beta^{(k)}(4f_{1/2,2/3}(s) - 1), & \text{if } 1/4 \le f_{1/2,2/3}(s) \le 1/2 \\ \gamma^{(k)}(2f_{1/2,2/3}(s) - 1), & \text{if } 1/2 \le f_{1/2,2/3}(s) \le 1 \end{cases}
\end{aligned}$$

Substituting the definitions of $f_{1/2,1/3}(s)$ and $f_{1/2,2/3}(s)$ and simplifying, we obtain

$$\begin{aligned}
&((\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha)^{(k)}(f_{1/2,1/3}(s)) \\
&= (\gamma \circ_{n-k} (\beta \circ_{n-k} \alpha))^{(k)}(f_{1/2,2/3}(s)) \\
&= \begin{cases} \alpha^{(k)}(3s), & \text{if } 0 \le s \le 1/3 \\ \beta^{(k)}(3s - 1), & \text{if } 1/3 \le s \le 2/3 \\ \gamma^{(k)}(3s - 2), & \text{if } 2/3 \le s \le 1, \end{cases}
\end{aligned}$$

which establishes (18).

We must argue that the source and target of (17) are correct. For the source,

$$\begin{aligned}
&\mathsf{src}(\mathsf{rpm}((\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha, f_{1/2,1/3}, k))^{(k)}(s) \\
&= \mathsf{rpm}((\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha, f_{1/2,1/3}, k))^{(1,k+1)}(0, s) \\
&= ((\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha)^{(k)}(s),
\end{aligned}$$

so $\mathsf{src}(\mathsf{rpm}((\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha, f_{1/2,1/3}, k)) = (\gamma \circ_{n-k} \beta) \circ_{n-k} \alpha$. Similarly,

$$\begin{aligned}
&\mathsf{tgt}(\mathsf{rpm}(\gamma \circ_{n-k} (\beta \circ_{n-k} \alpha), f^{\leftarrow}_{1/2,2/3}k,))^{(k)}(s) \\
&= \mathsf{rpm}(\gamma \circ_{n-k} (\beta \circ_{n-k} \alpha), f_{1/2,2/3}, k))^{(1,k+1)}(0, s) \\
&= \gamma \circ_{n-k} (\beta \circ_{n-k} \alpha)^{(k)}(s),
\end{aligned}$$

so $\mathsf{src}(\mathsf{rpm}(\gamma \circ_{n-k} (\beta \circ_{n-k} \alpha), f_{1/2,2/3}, k)) = \gamma \circ_{n-k} (\beta \circ_{n-k} \alpha)$. ◻

Lemma 8.2 (Identities). *If α is an n-cell and $1 \le k \le n$, then*

$$\alpha \circ_{n-k} \mathsf{id}^k \,\mathsf{src}^k \alpha \simeq_1 \alpha \qquad\qquad \mathsf{id}^k \,\mathsf{tgt}^k \alpha \circ_{n-k} \alpha \simeq_1 \alpha.$$

Proof. The composition $\alpha \circ_{n-k} \mathsf{id}^k \,\mathsf{src}^k \alpha$ exists, since

$$\begin{aligned}
(\mathsf{id}^k \,\mathsf{src}^k \alpha)^{(k)}(1) &= \mathsf{id}^{k-1} \,\mathsf{src}^k \,\mathsf{id}^k \,\mathsf{src}^k \alpha \\
&= \mathsf{id}^{k-1} \,\mathsf{src}^k \alpha = \alpha^{(k)}(0),
\end{aligned}$$

and $\mathsf{rpm}(\alpha, f_{0,1/2}, k)$ witnesses the equivalence with α:

$$\begin{aligned}
(\mathsf{src}\,\mathsf{rpm}(\alpha, f_{0,1/2}, k))^{(k)}(s) &= \mathsf{rpm}(\alpha, f_{0,1/2}, k)^{(1,k+1)}(0, s) \\
&= \alpha^{(k)}(f_{0,1/2}(0, s)) = \alpha^{(k)}(s),
\end{aligned}$$

so $\operatorname{src}\operatorname{rpm}(\alpha, f_{0,1/2}, 1) = \alpha$, and

$$\begin{aligned}(\operatorname{tgt}\operatorname{rpm}(\alpha, f_{0,1/2}, k))^{(k)}(s) &= \operatorname{rpm}(\alpha, f_{0,1/2}, k)^{(1,k+1)}(1, s) \\ &= \begin{cases}\alpha^{(k)}(0), & \text{if } 0 \le s \le 1/2 \\ \alpha^{(k)}(2s-1), & \text{if } 1/2 \le s \le 1\end{cases} \\ &= \begin{cases}(\mathrm{id}^k \operatorname{src}^k \alpha)^{(k)}(2s), & \text{if } 0 \le s \le 1/2 \\ \alpha^{(k)}(2s-1), & \text{if } 1/2 \le s \le 1\end{cases} \\ &= (\alpha \circ_{n-k} \mathrm{id}^k \operatorname{src}^k \alpha)^{(k)}(s),\end{aligned}$$

so $\operatorname{tgt}\operatorname{rpm}(\alpha, f_{0,1/2}, k) = \alpha \circ_{n-k} \mathrm{id}^k \operatorname{src}^k \alpha$.

The proof of $\mathrm{id}^k \operatorname{tgt}^k \alpha \circ_{n-k} \alpha \simeq_1 \alpha$ is symmetric, with witness $\operatorname{rpm}(\alpha, f_{1,1/2}, k)$. □

Lemma 8.3 (Inverse). *For any cell γ, $\gamma^{\leftarrow k} \circ_{n-k} \gamma$ exists and $\gamma^{\leftarrow k} \circ_{n-k} \gamma \simeq_1 \mathrm{id}^k \operatorname{src}^k \gamma$.*

Proof. We show that the reparameterization $\operatorname{rpm}(\gamma, u, k)$, where u is defined in (14), witnesses the desired equivalence. Substituting definitions gives

$$\begin{aligned}\operatorname{rpm}(\gamma, u, k)^{(1,k+1)}(t, s) &= \gamma^{(k)}(u(t, s)) \\ &= \begin{cases}\gamma^{(k)}(2ts), & \text{if } 0 \le s \le 1/2 \\ \gamma^{(k)}(2t(1-s)), & \text{if } 1/2 \le s \le 1.\end{cases}\end{aligned}$$

At $t = 0$, this is just $\gamma^{(k)}(0)$, so by Lemma 7.1,

$$\begin{aligned}(\operatorname{src}\operatorname{rpm}(\gamma, u, k))^{(k)}(s) &= \operatorname{rpm}(\gamma, u, k)^{(1,k+1)}(0, s) \\ &= \gamma^{(k)}(0) = \mathrm{id}^{k-1} \operatorname{src}^k \gamma,\end{aligned}$$

so $\operatorname{src}\operatorname{rpm}(\gamma, u, k) = \mathrm{id}^k \operatorname{src}^k \gamma$. At $t = 1$,

$$\begin{aligned}\operatorname{rpm}(\gamma, u, k)^{(1,k+1)}(1, s) &= \gamma^{(k)}(u(1, s)) \\ &= \begin{cases}\gamma^{(k)}(2s), & \text{if } 0 \le s \le 1/2 \\ \gamma^{(k)}(2(1-s)), & \text{if } 1/2 \le s \le 1\end{cases} \\ &= \begin{cases}\gamma^{(k)}(2s), & \text{if } 0 \le s \le 1/2 \\ \gamma^{(\leftarrow k)}(2s-1), & \text{if } 1/2 \le s \le 1\end{cases} \\ &= (\gamma^{\leftarrow k} \circ_{n-k} \gamma)^{(k)}(s),\end{aligned}$$

so

$$\begin{aligned}(\operatorname{tgt}\operatorname{rpm}(\gamma, u, k))^{(k)}(s) &= \operatorname{rpm}(\gamma, u, k)^{(1,k+1)}(1, s) \\ &= (\gamma^{\leftarrow k} \circ_{n-k} \gamma)^{(k)}(s),\end{aligned}$$

so $\operatorname{tgt}\operatorname{rpm}(\gamma, u, k) = \gamma^{\leftarrow k} \circ_{n-k} \gamma$. □

Lemma 8.5 (Exchange). *For all $1 \le k < m \le n$ and for all n-cells $\alpha, \beta, \gamma, \delta$ satisfying the requisite boundary conditions,*

$$(\alpha \circ_{n-k} \beta) \circ_{n-m} (\gamma \circ_{n-k} \delta) = (\alpha \circ_{n-m} \gamma) \circ_{n-k} (\beta \circ_{n-m} \delta),$$

therefore they are $\simeq_1$ equivalent as witnessed by $\mathrm{id}((\alpha \circ_{n-k} \beta) \circ_{n-m} (\gamma \circ_{n-k} \delta))$.

Proof. Expanding definitions, both expressions give

$$\begin{aligned}&((\alpha \circ_{n-k} \beta) \circ_{n-m} (\gamma \circ_{n-k} \delta))^{(k,m)}(x, y) \\ &= ((\alpha \circ_{n-m} \gamma) \circ_{n-k} (\beta \circ_{n-m} \delta))^{(k,m)}(x, y) \\ &= \begin{cases}\alpha^{(k,m)}(2x, 2y), & \text{if } 0 \le x \le 1/2 \text{ and } 0 \le y \le 1/2 \\ \beta^{(k,m)}(2x-1, 2y), & \text{if } 1/2 \le x \le 1 \text{ and } 0 \le y \le 1/2 \\ \gamma^{(k,m)}(2x, 2y-1), & \text{if } 0 \le x \le 1/2 \text{ and } 1/2 \le y \le 1 \\ \delta^{(k,m)}(2x-1, 2y-1), & \text{if } 1/2 \le x \le 1 \text{ and } 1/2 \le y \le 1.\end{cases}\end{aligned}$$

□

Lemma 8.6 (Congruence of $\circ_{n-k}$). *Let α, β, γ be n-cells. If $\operatorname{tgt}^k \alpha = \operatorname{tgt}^k \beta = \operatorname{src}^k \gamma$ and $\alpha \simeq_1 \beta$, then $\gamma \circ_{n-k} \alpha \simeq_1 \gamma \circ_{n-k} \beta$. Symmetrically, if $\operatorname{src}^k \alpha = \operatorname{src}^k \beta = \operatorname{tgt}^k \gamma$ and $\alpha \simeq_1 \beta$, then $\alpha \circ_{n-k} \gamma \simeq_1 \beta \circ_{n-k} \gamma$.*

Proof. We prove only the former statement, as the latter is symmetrical. The case $k = 0$ is trivial, so assume $k \geq 1$. Let φ witness $\alpha \simeq_1 \beta$, so $\operatorname{src} \varphi = \alpha$ and $\operatorname{tgt} \varphi = \beta$. Then $\operatorname{tgt}^{k+1} \varphi = \operatorname{tgt}^k \operatorname{tgt} \varphi = \operatorname{tgt}^k \beta$, and also $\operatorname{tgt}^{k+1} \varphi = \operatorname{tgt}^k \operatorname{tgt} \varphi = \operatorname{tgt}^k \operatorname{src} \varphi = \operatorname{tgt}^k \alpha$, and both of these are equal to $\operatorname{src}^k \gamma$. Now let $\psi = \operatorname{id} \gamma$. Then $\operatorname{src} \psi = \operatorname{tgt} \psi = \gamma$, thus $\operatorname{src}^{k+1} \psi = \operatorname{src}^k \gamma$, so $\psi \circ_{n-k} \varphi$ exists and is the desired witness:

$$\operatorname{src}(\psi \circ_{n-k} \varphi) = \operatorname{src} \psi \circ_{n-k} \operatorname{src} \varphi = \gamma \circ_{n-k} \alpha$$
$$\operatorname{tgt}(\psi \circ_{n-k} \varphi) = \operatorname{tgt} \psi \circ_{n-k} \operatorname{tgt} \varphi = \gamma \circ_{n-k} \beta. \qquad \square$$

Lemma 8.7. *The following are equivalent:*

- *α and $\beta^{\leftarrow k} \circ_{n-k} \alpha$ are in C_n and $\beta^{\leftarrow k} \circ_{n-k} \alpha \simeq_1 \operatorname{id}^k \operatorname{src}^k \alpha$;*
- *α and β are in C_n and $\alpha \simeq_1 \beta$.*

Proof. For any $\gamma \simeq_1 \delta$ witnessed by φ, we have

$$\operatorname{src} \gamma = \operatorname{src}^2 \varphi = \operatorname{src} \operatorname{tgt} \varphi = \operatorname{src} \delta \qquad\qquad \operatorname{tgt} \gamma = \operatorname{tgt} \operatorname{src} \varphi = \operatorname{tgt}^2 \varphi = \operatorname{tgt} \delta,$$

and it follows that $\operatorname{src}^k \gamma = \operatorname{src}^k \delta$ and $\operatorname{tgt}^k \gamma = \operatorname{tgt}^k \delta$ for all $k \geq 1$.

Assume first that α and $\beta^{\leftarrow k} \circ_{n-k} \alpha$ are in C_n and $\beta^{\leftarrow k} \circ_{n-k} \alpha \simeq_1 \operatorname{id}^k \operatorname{src}^k \alpha$. Using (9), these assumptions imply that

$$\operatorname{src}^k \alpha = \operatorname{tgt}^k \operatorname{id}^k \operatorname{src}^k \alpha = \operatorname{tgt}^k(\beta^{\leftarrow k} \circ_{n-k} \alpha) = \operatorname{tgt}^k \beta^{\leftarrow k} = \operatorname{src}^k \beta$$
$$\operatorname{tgt}^k \alpha = \operatorname{src}^k \beta^{\leftarrow k} = \operatorname{tgt}^k \beta.$$

Then

$$\begin{aligned}
\alpha &\simeq_1 \operatorname{id}^k \operatorname{tgt}^k \alpha \circ_{n-k} \alpha && \text{Lemma 8.2}\\
&= \operatorname{id}^k \operatorname{tgt}^k \beta \circ_{n-k} \alpha && \text{assumption}\\
&\simeq_1 (\beta \circ_{n-k} \beta^{\leftarrow k}) \circ_{n-k} \alpha && \text{Lemma 8.3}\\
&\simeq_1 \beta \circ_{n-k} (\beta^{\leftarrow k} \circ_{n-k} \alpha) && \text{Lemma 8.1}\\
&\simeq_1 \beta \circ_{n-k} \operatorname{id}^k \operatorname{src}^k \alpha && \text{Lemma 8.6 and assumption}\\
&= \beta \circ_{n-k} \operatorname{id}^k \operatorname{src}^k \beta && \text{assumption}\\
&\simeq_1 \beta && \text{Lemma 8.2.}
\end{aligned}$$

Then β exists and is $\simeq_1$-equivalent to α. Conversely, if α and β are in C_n and $\alpha \simeq_1 \beta$, then $\operatorname{src}^k \alpha = \operatorname{src}^k \beta$ and $\operatorname{tgt}^k \alpha = \operatorname{tgt}^k \beta$, so

$$\begin{aligned}
\operatorname{id}^k \operatorname{src}^k \alpha &= \operatorname{id}^k \operatorname{src}^k \beta && \text{assumption}\\
&\simeq_1 \beta^{\leftarrow k} \circ_{n-k} \beta && \text{Lemma 8.3}\\
&\simeq_1 \beta^{\leftarrow k} \circ_{n-k} \alpha && \text{Lemma 8.6 and assumption}
\end{aligned}$$

so $\beta^{\leftarrow k} \circ_{n-k} \alpha$ exists and is $\simeq_1$-equivalent to $\operatorname{id}^k \operatorname{src}^k \alpha$. $\square$

Lemma 8.8. *$\alpha \simeq_1 \beta$ if and only if*

$$\exists \varphi : \alpha \to \beta \; \exists \psi : \beta \to \alpha \;\; \psi \circ_{n-1} \varphi \simeq_1 \operatorname{id} \alpha \;\wedge\; \varphi \circ_{n-1} \psi \simeq_1 \operatorname{id} \beta.$$

Proof. Suppose $\alpha \simeq_1 \beta$ as witnessed by φ. Then $\operatorname{src} \varphi = \alpha$, $\operatorname{tgt} \varphi = \beta$, and φ is invertible, therefore $\varphi^{\leftarrow 1}$ exists. Then $\varphi : \alpha \to \beta$, $\varphi^{\leftarrow 1} : \beta \to \alpha$, and by Lemma 8.3,

$$\varphi^{\leftarrow 1} \circ_{n-1} \varphi \simeq_1 \operatorname{id} \operatorname{src} \varphi = \operatorname{id} \alpha$$
$$\varphi \circ_{n-1} \varphi^{\leftarrow 1} \simeq_1 \operatorname{id} \operatorname{src} \varphi^{\leftarrow} = \operatorname{id} \beta.$$

Conversely, suppose $\varphi : \alpha \to \beta$, $\psi : \beta \to \alpha$, $\psi \circ_{n-1} \varphi \simeq_1 \operatorname{id} \alpha$, and $\varphi \circ_{n-1} \psi \simeq_1 \operatorname{id} \beta$. Because $\psi \circ_{n-1} \varphi \simeq_1 \operatorname{id} \alpha = \operatorname{id} \operatorname{src} \varphi$, we have that $\varphi \in C_{n+1}$, $(\psi^{\leftarrow 1})^{\leftarrow 1} \circ_{n-1} \varphi \in C_{n+1}$, and $(\psi^{\leftarrow 1})^{\leftarrow 1} \circ_{n-1} \varphi \simeq_1 \operatorname{id} \operatorname{src} \varphi$. By Lemma 8.7, φ and $\psi^{\leftarrow 1}$ are in C_{n+1} and $\varphi \simeq_1 \psi^{\leftarrow 1}$, thus ψ is invertible and provides a witness for $\alpha \simeq_1 \beta$. $\square$

E THE COMPACT-OPEN TOPOLOGY

Let X and K be topological spaces with K compact and Hausdorff. (In our applications, K is the hypercube $[0,1]^n$.) Every compact Hausdorff space is regular, which means that for every closed set C and point $x \notin C$, there exist disjoint open sets U, V such that $x \in U$ and $C \subseteq V$.

Let X^K be the space of continuous functions $K \to X$ with the compact-open topology. This topology is generated by subbasic open sets $U^C = \{f \in X^K \mid C \subseteq f^{-1}(U)\}$ for $C \subseteq K$ closed (thus compact) and $U \subseteq X$ open.

Lemma E.1. *Let A, B, and K be topological spaces with K compact and Hausdorff. Let $h : A \times K \to B$ be a function and let $h' : A \to K \to B$ be its curried form $h'(a)(x) = h(a, x)$. The following are equivalent:*

(i) *h is continuous with respect to the product topology on $A \times K$;*
(ii) *$h' : A \to B^K$ (that is, for all $a \in A$, $h'(a) : K \to B$ is continuous) and h' is continuous with respect to the compact-open topology on B^K.*

Thus $- \times K$ is left adjoint to $-^K$.

Proof. (i) ⟹ (ii). Suppose $h : A \times K \to B$ is continuous. We first show that for all $a \in A$, $h'(a) : K \to B$ is continuous. For any open set $U \subseteq B$, by elementary set theory, $h'(a)^{-1}(U) = \{x \in K \mid (a, x) \in h^{-1}(U)\}$. This is a section of an open set in a product space, therefore is open.

To show $h' : A \to B^K$ is continuous, let U^C be a subbasic open set in B^K. By elementary set theory, $a \in h'^{-1}(U^C)$ iff $\{a\} \times C \subseteq h^{-1}(U)$. Since $h^{-1}(U)$ is open in $A \times K$, for all $x \in C$ there exist basic open sets $U_x \times V_x \subseteq h^{-1}(U)$ such that $a \in U_x$ and $x \in V_x$. By compactness, there is a finite set $F \subseteq C$ such that $C \subseteq \bigcup_{x \in F} V_x$ and $a \in \bigcap_{y \in F} U_x$, thus

$$\{a\} \times C \subseteq \bigcap_{y \in F} U_y \times \bigcup_{x \in F} V_x = \bigcup_{x \in F} \bigcap_{y \in F} U_y \times V_x \subseteq h^{-1}(U).$$

Again by elementary set theory, $a \in \bigcap_{y \in F} U_y \subseteq h'^{-1}(U^C)$.

(ii) ⟹ (i). Suppose $h' : A \to B^K$ is continuous. Let $\varepsilon_B : B^K \times K \to B$ be the application function $\varepsilon_B(f, x) = f(x)$. We first show that ε_B is continuous. For any $U \subseteq B$ open, $\varepsilon_B^{-1}(U) = \{(f, x) \mid f(x) \in U\}$. If $(f, x) \in \varepsilon_B^{-1}(U)$, then $f : K \to B$ is continuous, so $f^{-1}(U)$ is open and $x \in f^{-1}(U)$. Since K is regular, there exists an open set V with closure C such that $x \in V \subseteq C \subseteq f^{-1}(U)$. Then $f \in U^C$ since $C \subseteq f^{-1}(U)$, so $(f, x) \in U^C \times V \subseteq U^C \times C \subseteq \varepsilon_B^{-1}(U)$, and $U^C \times V$ is open, therefore ε_B is continuous.

Now for all $(a, x) \in A \times K$,

$$(\varepsilon_B \circ (h' \times \mathsf{Id}_K))(a, x) = \varepsilon_B((h' \times \mathsf{Id}_K)(a, x)) = \varepsilon_B(h'(a), x) = h'(a)(x) = h(a, x),$$

so $h = \varepsilon_B \circ (h' \times \mathsf{Id}_K)$. This is a composition of continuous functions, so h is continuous. □